

# Differential Equations and Linear Algebra

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All errors, typographical and substantive, and other offenses, are entirely my own.

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## 8 - Fourier and Laplace Transforms

### 8.1 - Fourier Series

#### Question: 8.1.1

(a) To prove that  $\cos(nx)$  is orthogonal to  $\cos(kx)$  when  $k \neq n$ , use the formula  $(\cos(nx))(\cos(kx)) = \frac{1}{2} \cos((n+k)x) + \frac{1}{2} \cos((n-k)x)$ . Integrate from  $x = 0$  to  $x = \pi$ . What is  $\int \cos^2(kx) dx$ ?

(b) From 0 to  $\pi$ ,  $\cos(x)$  is **not** orthogonal to  $\sin(x)$ . The period has to be  $2\pi$ :

Find  $\int_0^\pi \sin(x) \cos(x) dx$  and  $\int_{-\pi}^\pi \sin(x) \cos(x) dx$  and

$$\int_0^{2\pi} \sin(x) \cos(x) dx.$$

(a) For  $k \neq n$ , integrate

$$\begin{aligned} \int_0^\pi \cos(nx) \cos(kx) dx &= \int_0^\pi \left[ \frac{1}{2} \cos((n+k)x) + \frac{1}{2} \cos((n-k)x) \right] dx \\ &= \frac{1}{2} \left[ \frac{1}{n+k} \sin((n+k)x) + \frac{1}{n-k} \sin((n-k)x) \right] \Big|_0^\pi \\ &= 0 \end{aligned}$$

When equality holds, we have

$$\int_0^\pi \cos^2(kx) dx = \int_0^\pi \left[ \frac{1}{2} \cos(2kx) + \frac{1}{2} \right] dx = \frac{\pi}{2}$$

(b) I think the textbook has a typo. The integrand should be  $\sin(2x) \cos(x)$ .

$$\begin{aligned} \int_0^\pi \sin(2x) \cos(x) dx &= \int_0^\pi 2 \sin(x) \cos^2(x) dx \\ &= -\frac{2}{3} \cos^3(x) \Big|_0^\pi \\ &= \frac{4}{3} \end{aligned}$$

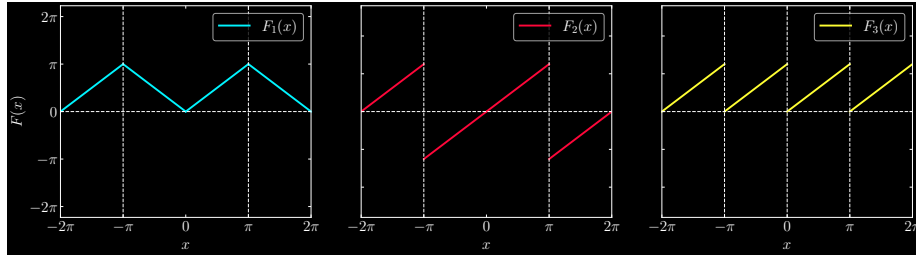
When we set the integral to the two other sets of bounds, we find

$$-\frac{2}{3} \cos^3(x) \Big|_{-\pi}^\pi = -\frac{2}{3} \cos^3(x) \Big|_0^{2\pi} = 0$$

**Question: 8.1.2**

Suppose  $F(x) = x$  for  $0 \leq x \leq \pi$ . Draw graphs for  $-2\pi \leq x \leq 2\pi$  to show three extensions of  $F$ : a  $2\pi$ -periodic even function and a  $2\pi$ -periodic odd function and a  $\pi$ -periodic function.

Graph each extension  $F_1$ ,  $F_2$ , and  $F_3$  as

**Question: 8.1.3**

Find the Fourier series on  $-\pi \leq x \leq \pi$  for

- (a)  $f_1(x) = \sin^3(x)$ , an odd function (sine series, only two terms)
- (b)  $f_2(x) = |\sin(x)|$ , an even function (cosine series)
- (c)  $f_3(x) = x$  for  $-\pi \leq x \leq \pi$  (sine series with jump at  $x = \pi$ )

- (a) The Fourier coefficients are given by

$$b_k = \frac{2}{\pi} \int_{-\pi}^{\pi} \sin^3(x) \sin(kx) dx$$

The simplest approach is to write the identity  $\sin(x) = (e^{ix} - e^{-ix})/2i$ . Then we can calculate

$$\begin{aligned} \sin^3(x) &= -\frac{1}{8i} (e^{ix} - e^{-ix})^3 \\ &= -\frac{1}{8i} (e^{3ix} - e^{-3ix} - 3(e^{ix} - e^{-ix})) \\ &= -\frac{1}{4} (\sin(3x) - 3\sin(x)) \end{aligned}$$

which is a trigonometric identity that itself is also the Fourier series for  $\sin^3(x)$ , and therefore  $b_1 = 3/4$  and  $b_2 = -1/4$ .

- (b) Note that the absolute value means

$$|\sin(x)| = \begin{cases} \sin(x) & 0 \leq x \leq \pi \\ -\sin(x) & -\pi \leq x < 0 \end{cases}$$

So calculating  $a_0$  yields

$$a_0 = \frac{1}{2\pi} \left[ \int_0^{\pi} \sin(x) dx + \int_{-\pi}^0 (-\sin(x)) dx \right] = \frac{2}{\pi}$$



Deriving the Fourier coefficients, we find

$$\begin{aligned} a_k &= \frac{2}{\pi} \int_0^\pi \sin(x) \cos(kx) \, dx \\ &= -\frac{1}{\pi} \left[ \frac{\cos(1-k)x}{1-k} + \frac{\cos(1+k)x}{1+k} \right] \Big|_0^\pi \\ &= -\frac{1}{\pi(k^2-1)} [(1+k) \cos(1-k)x + (1-k) \cos(1+k)x] \Big|_0^\pi \end{aligned}$$

Now, when  $k$  is odd, this expression vanishes. Otherwise, for even  $k$  it reduces to

$$\begin{aligned} &= \frac{1}{\pi(k^2-1)} [-2(1+k) - 2(1-k)] \\ &= \frac{4}{\pi(1-k^2)} \end{aligned}$$

To cleanly put this in a summation, let  $k = 2n$ . Then

$$|\sin(x)| = \frac{2}{\pi} + \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\cos(2nx)}{1-4n^2}, \quad -\pi \leq x \leq \pi$$

(c) The Fourier coefficients are

$$\begin{aligned} b_k &= \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin(kx) \, dx \\ &= \frac{1}{\pi} \left[ \frac{\sin(kx)}{k^2} - \frac{x \cos(kx)}{k} \right] \Big|_{-\pi}^{\pi} \end{aligned}$$

which is  $2/k$  when  $k$  is odd and  $-2/k$  when  $k$  is even. By symmetry,  $b_0 = 0$ . The Fourier sine series is

$$x = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin(nx)$$

#### Question: 8.1.4

Find the complex Fourier series  $e^x = \sum c_k e^{ikx}$  on the interval  $-\pi \leq x \leq \pi$ . The even part of a function is  $\frac{1}{2}(f(x) + f(-x))$ , so that  $f_{\text{even}}(x) = f_{\text{even}}(-x)$ . Find the cosine series for  $f_{\text{even}}$  and the sine series for  $f_{\text{odd}}$ . Notice the jump at  $x = \pi$ .

First, we calculate the coefficients to the complex Fourier series as

$$\begin{aligned} c_k &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^x e^{-ikx} \, dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{(1-ik)x} \, dx \\ &= \frac{1}{2\pi} \frac{e^{(1-ik)x}}{1-ik} \Big|_{-\pi}^{\pi} \\ &= \frac{1}{2\pi(1-ik)} [e^{(1-ik)\pi} - e^{-(1-ik)\pi}] \end{aligned}$$

Next, decompose the exponential function into the well-known identity

$$e^x = \underbrace{\frac{1}{2}(e^x + e^{-x})}_{\text{even}} + \underbrace{\frac{1}{2}(e^x - e^{-x})}_{\text{odd}} = \cosh(x) + \sinh(x)$$

The Fourier cosine series for the even component is

$$\begin{aligned} a_k &= \int_{-\pi}^{\pi} \frac{1}{2}(e^x + e^{-x}) \cos(kx) \, dx \\ &= \frac{2}{\pi} \int_0^{\pi} \frac{1}{2}(e^x + e^{-x}) \cos(kx) \, dx \\ &= \frac{1}{\pi} \left[ \frac{e^x(\cos(kx) + k \sin(kx))}{1+k^2} + \frac{e^{-x}(-\cos(kx) + k \sin(kx))}{1+k^2} \right] \Big|_0^{\pi} \\ &= \frac{(-1)^k}{\pi} \left( \frac{e^{\pi} - e^{-\pi}}{1+k^2} \right) \\ &= (-1)^k \frac{2}{\pi(1+k^2)} \sinh(\pi) \end{aligned}$$

And for the Fourier sine series for the odd component, we have

$$\begin{aligned} b_k &= \frac{1}{\pi} \int_{-\pi}^{\pi} \frac{1}{2}(e^x - e^{-x}) \sin(kx) \, dx \\ &= \frac{2}{\pi} \int_0^{\pi} \frac{1}{2}(e^x - e^{-x}) \sin(kx) \, dx \\ &= \frac{1}{\pi} \left[ \frac{e^x(\sin(kx) - k \cos(kx))}{1+k^2} - \frac{e^{-x}(-\sin(kx) - k \cos(kx))}{1+k^2} \right] \Big|_0^{\pi} \\ &= (-1)^{k+1} \frac{k}{\pi} \left( \frac{e^{\pi} - e^{-\pi}}{1+k^2} \right) \\ &= (-1)^{k+1} \frac{2k}{\pi(1+k^2)} \sinh(\pi) \end{aligned}$$

Additionally, the average is

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^x \, dx = \frac{1}{2\pi} (e^{\pi} - e^{-\pi}) = \frac{1}{\pi} \sinh(\pi)$$

The full Fourier series for our  $F(x)$  is then

$$e^x = \frac{1}{\pi} \sinh(\pi) + \sum_{k=1}^{\infty} (-1)^k \frac{2}{\pi(1+k^2)} \sinh(\pi) \cos(kx) - \sum_{k=1}^{\infty} (-1)^k \frac{2k}{\pi(1+k^2)} \sinh(\pi) \sin(kx)$$

**Question: 8.1.5**

From the energy formula (21), the square wave sine coefficients satisfy

$$\pi(b_1^2 + b_2^2 + \cdots) = \int_{-\pi}^{\pi} |SW(x)|^2 dx = \int_{-\pi}^{\pi} 1 dx = 2\pi$$

Substitute the numbers  $b_k$  from equation (8) to find that  $\pi^2 = 8\left(1 + \frac{1}{9} + \frac{1}{25} + \cdots\right)$ .

The coefficients  $b_k$  are

$$b_k = \frac{2}{\pi} \left\{ \frac{2}{1}, \frac{2}{3}, \frac{2}{5}, \cdots \right\}$$

Substituting gives us

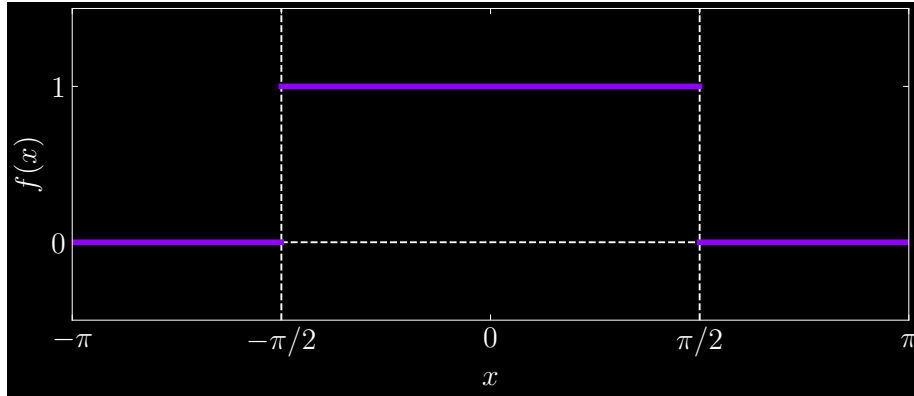
$$16\pi \left( \frac{1}{\pi^2} + \frac{1}{9\pi^2} + \frac{1}{25\pi^2} + \cdots \right) = 2\pi \implies \pi^2 = 8 \left( 1 + \frac{1}{9} + \frac{1}{25} + \cdots \right)$$

**Question: 8.1.6**

If a square pulse is centered at  $x = 0$  to give

$$f(x) = 1 \quad \text{for } |x| < \frac{\pi}{2}, \quad f(x) = 0 \quad \text{for } \frac{\pi}{2} < |x| < \pi,$$

draw its graph and find its Fourier coefficients  $a_k$  and  $b_k$ .



On visual inspection, it is apparent that  $f(x)$  is an even function. From that we can immediately conclude that all  $b_k = 0$ . Then we calculate

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} dx = \frac{1}{2}$$

$$\begin{aligned}
 a_k &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(kx) \, dx \\
 &= \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} \cos(kx) \, dx \\
 &= \frac{2}{k\pi} \sin\left(\frac{k\pi}{2}\right)
 \end{aligned}$$

giving us complete Fourier series

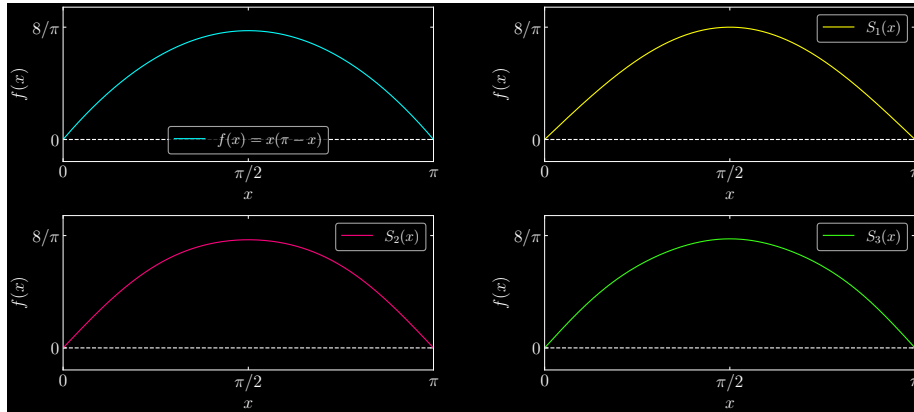
$$f(x) = \frac{1}{2} + \sum_{k=1}^{\infty} \frac{2}{k\pi} \sin\left(\frac{k\pi}{2}\right) \cos(kx)$$

### Question: 8.1.7

Plot the first three partial sums and the function  $x(\pi - x)$ :

$$x(\pi - x) = \frac{8}{\pi} \left( \frac{\sin(x)}{1} + \frac{\sin(3x)}{27} + \frac{\sin(5x)}{125} + \cdots \right), \quad 0 < x < \pi$$

Why is  $1/k^3$  the decay rate for this function? What is its second derivative?



The second derivative is  $f'' = -2$ . One analysis here is we must integrate thrice to derive our Fourier coefficient: twice for the  $x^2$  term, and once more for the sine. This generates  $1/k^3$ .

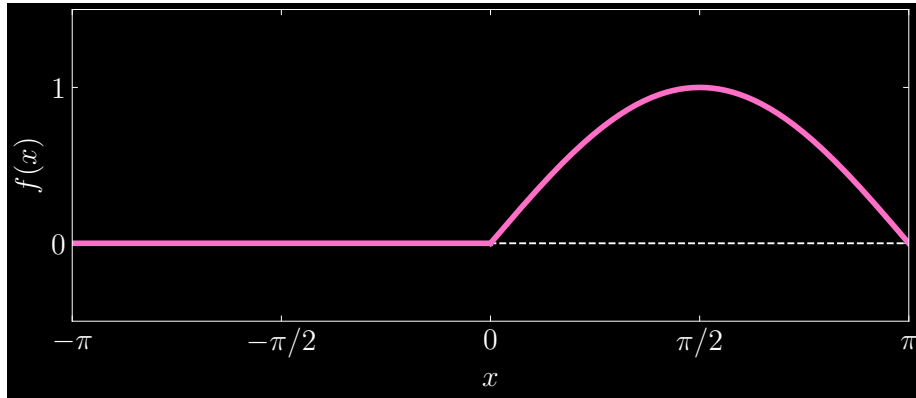
Another way to think about the decay rate is as follows:  $1/k$  corresponds to jumps in the function itself (step functions),  $1/k^2$  is the jump in the first derivative of the ramp function, and  $1/k^3$  for the jump in the second derivative. The argument here, then, requires us to consider the periodic odd extension of  $f$ , namely the case where

$$f = x\pi + x^2, \quad -\pi < x < 0$$

In this “branch”, the second derivative is  $+2$ . At  $x = 0$ , we have a discontinuity in the second derivative: the jump from  $+2$  to  $-2$ .

**Question: 8.1.8**

Sketch the  $2\pi$ -periodic half wave with  $f(x) = \sin(x)$  for  $0 < x < \pi$  and  $f(x) = 0$  for  $-\pi < x < 0$ . Find its Fourier series.



The Fourier series is calculated as

$$\begin{aligned}
 a_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \, dx \\
 &= \frac{1}{2\pi} \int_0^{\pi} \sin(x) \, dx \\
 &= \frac{1}{\pi} \\
 a_k &= \frac{1}{\pi} \int_0^{\pi} \sin(x) \cos(kx) \, dx \\
 &= \frac{1}{\pi} \left[ \frac{\cos((1-k)x)}{2(1-k)} - \frac{\cos((1+k)x)}{2(1+k)} \right] \Big|_0^{\pi} \\
 &= \frac{1}{\pi} \left[ \frac{\cos(k\pi) + 1}{2(1-k)} + \frac{\cos(k\pi) + 1}{2(1+k)} \right] \\
 \Rightarrow a_n &= \frac{1}{\pi} \left[ \frac{1}{1-2n} + \frac{1}{1+2n} \right] \quad (\text{substitute } k = 2n; \text{ non-zero only for even } k) \\
 &= \frac{2}{\pi(1-4n^2)} \\
 b_k &= \frac{1}{\pi} \int_0^{\pi} \sin(x) \sin(kx) \, dx \\
 &= \frac{1}{\pi} \left[ \frac{x}{2} - \frac{\sin(2x)}{4} \right] \Big|_0^{\pi} \\
 &= \frac{1}{2}
 \end{aligned}$$

which gives us

$$f(x) = \frac{1}{\pi} + \sum_{n=1}^{\infty} \frac{2}{\pi(1-4n^2)} + \frac{\sin(x)}{2}$$

**Question: 8.1.9**

Suppose  $G(x)$  has period  $2L$  instead of  $2\pi$ . Then  $G(x + 2L) = G(x)$ . Integrals go from  $-L$  to  $L$  or from  $0$  to  $2L$ . The Fourier formulas change by a factor  $\pi/L$ :

The coefficients in

$$G(x) = \sum_{-\infty}^{\infty} C_k e^{ik\pi x/L} \text{ are } C_k = \frac{1}{2L} \int_{-L}^L G(x) e^{-ik\pi x/L} dx.$$

Derive this formula for  $C_k$ : Multiply the first equation for  $G(x)$  by \_\_\_\_\_ and integrate both sides. Why is the integral on the right side equal to  $2LC_k$ ?

Multiply the first equation by  $e^{-ik\pi x/L}$ .

$$G(x) e^{-ik\pi x/L} = e^{-ik\pi x/L} \sum_{-\infty}^{\infty} C_k e^{ik\pi x/L}$$

By orthogonality of the complex exponentials, after integrating, we are left only with

$$\int_{-L}^L G(x) e^{-ik\pi x/L} dx = \int_{-L}^L C_k dx = 2LC_k$$

**Question: 8.1.10**

For  $G_{\text{even}}$ , use Problem 9 to find the cosine coefficient  $A_k$  from  $(C_k + C_{-k})/2$ :

$$G_{\text{even}}(x) = \sum_0^{\infty} A_k \cos\left(\frac{k\pi x}{L}\right) \text{ has } A_k = \frac{1}{L} \int_0^L G_{\text{even}}(x) \cos\left(\frac{k\pi x}{L}\right) dx.$$

$G_{\text{even}}$  is  $\frac{1}{2}(G(x) + G(-x))$ . Exception for  $A_0 = C_0$ : Divide by  $2L$  instead of  $L$ .

Calculate

$$\begin{aligned} \frac{c_k + c_{-k}}{2} &= \frac{1}{4L} \left[ \int_{-L}^L G(x) e^{-ik\pi x/L} dx + \int_{-L}^L G(x) e^{ik\pi x/L} dx \right] \\ &= \frac{1}{2L} \int_{-L}^L G(x) \cos\left(\frac{k\pi x}{L}\right) dx \\ &= \frac{1}{2L} \int_{-L}^L \left[ \underbrace{\frac{1}{2}(G(x) + G(-x))}_{\text{even}} + \underbrace{\frac{1}{2}(G(x) - G(-x))}_{\text{odd}} \right] \cos\left(\frac{k\pi x}{L}\right) dx \\ &= \frac{1}{2L} \int_{-L}^L G_{\text{even}}(x) \cos\left(\frac{k\pi x}{L}\right) dx \\ &= \frac{1}{L} \int_0^L G_{\text{even}}(x) \cos\left(\frac{k\pi x}{L}\right) dx \\ &= A_k \end{aligned}$$

Recall that in the third line, since an odd function times an even function is odd, the second integral vanishes over the symmetric bounds. This leaves only the even component of  $G(x)$ .

#### Question: 8.1.11

Problem 10 tells us that  $\mathbf{a}_k = \frac{1}{2}(\mathbf{c}_k + \mathbf{c}_{-k})$  on the usual interval from 0 to  $\pi$ . Find a similar formula for  $b_k$  from  $c_k$  and  $c_{-k}$ . In the reverse direction, find the complex coefficient  $c_k$  in  $F(x) = \sum c_k e^{ikx}$  from the real coefficients  $a_k$  and  $b_k$ .

A point of clarification is warranted. Problem 8.1.10 calculates the Fourier coefficient for the *even extension* of  $G(x)$ . We can do the same for the *odd* extension of  $G(x)$ , which involves reworking the derivation in the previous problem to ensure that the sine term survives. The way to do so is to begin from  $(c_k - c_{-k})/2$ . Doing so leaves us with

$$\frac{c_k - c_{-k}}{2} = -\frac{i}{2L} \int_{-L}^L G_{\text{odd}}(x) \sin\left(\frac{k\pi x}{L}\right) dx$$

Multiply by  $i$  to conclude

$$i\left(\frac{c_k - c_{-k}}{2}\right) = \frac{1}{2L} \int_{-L}^L G_{\text{odd}}(x) \sin\left(\frac{k\pi x}{L}\right) dx = B_k$$

Now, we consider the original function  $G(x)$  (with the definition in Problem 8.1.9). Fix  $k$  and write

$$\begin{aligned} c_k &= \frac{1}{2L} \int_{-L}^L G(x) e^{-ik\pi x/L} dx \\ &= \frac{1}{2L} \int_{-L}^L G(x) \left[ \cos\left(\frac{k\pi x}{L}\right) - i \sin\left(\frac{k\pi x}{L}\right) \right] dx \\ &= \frac{1}{2}(a_k - ib_k) \end{aligned}$$

Analogously conclude that

$$c_{-k} = \frac{1}{2}(a_k + ib_k)$$

Two equations for two unknowns gives us

$$\begin{aligned} a_k &= c_k + c_{-k} \\ b_k &= i(c_k - c_{-k}) \end{aligned}$$

which are the great and elegant formulae for relating the complex and trigonometric Fourier coefficients for  $G(x)$ .

**Question: 8.1.12**

Find the solution to Laplace's equation with  $u_0 = \theta$  on the boundary. Why is this the imaginary part of  $2(z - z^2/2 + z^3/3 \cdots) = 2\log(1 + z)$ ? Confirm that on the unit circle  $z = e^{i\theta}$ , the imaginary part of  $2\log(1 + z)$  agrees with  $\theta$ .

Since  $u_0(\theta) = \theta$  is an odd function, we calculate the sine series as

$$\begin{aligned} b_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} \theta \, d\theta \\ &= 0 \\ b_k &= \frac{1}{\pi} \int_{-\pi}^{\pi} \theta \sin(k\theta) \, d\theta \\ &= \frac{1}{\pi} \left[ \frac{\sin(k\theta)}{k^2} - \frac{\theta \cos(k\theta)}{k} \right] \Big|_{-\pi}^{\pi} \\ &= \frac{2}{k} (-1)^{k+1} \end{aligned}$$

The solution to Laplace's equation is

$$u(r, \theta) = \frac{2}{1} r \sin(\theta) - \frac{2}{2} r^2 \sin(2\theta) + \frac{2}{3} r^3 \sin(3\theta) - \cdots = \sum_{k=1}^{\infty} \frac{2}{k} (-1)^{k+1} r^k \sin(k\theta)$$

Let  $z = re^{i\theta}$ . Then this solution indeed corresponds to  $\text{Im}[2\log(1 + z)]$ .

**Question: 8.1.13**

If the boundary condition for Laplace's equation is  $u_0 = 1$  for  $0 < \theta < \pi$  and  $u_0 = 0$  for  $-\pi < \theta < 0$ , find the Fourier series solution  $u(r, \theta)$  inside the unit circle. What is  $u$  at the origin  $r = 0$ ?

We must calculate the complete Fourier series, which is

$$\begin{aligned} a_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} u_0 \, d\theta \\ &= \frac{1}{2\pi} \int_0^{\pi} d\theta \\ &= \frac{1}{2} \\ a_k &= \frac{1}{\pi} \int_0^{\pi} \cos(k\theta) \, d\theta \\ &= 0 \\ b_k &= \frac{1}{\pi} \int_0^{\pi} \sin(k\theta) \, d\theta \\ &= \frac{1}{\pi} \left[ \frac{(-1)^{k+1} + 1}{k} \right] \end{aligned}$$



This vanishes for even  $k$ , so we should write

$$b_n = \frac{2}{\pi(2n-1)}$$

The solution to Laplace's equation is

$$u(r, \theta) = \frac{1}{2} + \sum_{n=1}^{\infty} \frac{2}{\pi(2n-1)} r^{2n-1} \sin((2n-1)\theta)$$

#### Question: 8.1.14

With boundary values  $u_0(\theta) = 1 + \frac{1}{2}e^{i\theta} + \frac{1}{4}e^{2i\theta} + \dots$ , what is the Fourier series solution to Laplace's equation in the circle? Sum this geometric series.

Multiply each term by the corresponding power of  $r$  to get

$$u(r, \theta) = \sum_{n=0}^{\infty} \left( \frac{1}{2} r e^{i\theta} \right)^n$$

For this infinite sum, we have closed-form expression

$$u(r, \theta) = \frac{1}{1 - \frac{1}{2} r e^{i\theta}}$$

#### Question: 8.1.15

- Verify that the fraction in Poisson's formula (30) satisfies Laplace's equation.
- Find the response  $u(r, \theta)$  to an impulse at  $x = 0$ ,  $y = 1$  (where  $\theta = \frac{\pi}{2}$ ).

- In  $(r, \theta)$  coordinates, Laplace's equation is

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0$$

Note that the Poisson's equation has, for the purposes of differentiation, a more legible summation form

$$u(r, \theta) = \frac{1}{2\pi} + \frac{1}{\pi} \sum_{n=1}^{\infty} r^n \cos(n\theta)$$

Each term is calculated as

$$\begin{aligned} \frac{\partial^2 u}{\partial r^2} &= \frac{1}{\pi} \sum_{n=1}^{\infty} n(n-1) r^{n-2} \cos(n\theta) \\ \frac{1}{r} \frac{\partial u}{\partial r} &= \frac{1}{\pi} \sum_{n=1}^{\infty} n r^{n-2} \cos(n\theta) \\ \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} &= -\frac{1}{\pi} \sum_{n=1}^{\infty} n^2 r^{n-2} \cos(n\theta) \end{aligned}$$

which, expectedly, vanish upon summation.

- (b) Think of the impulse as undergoing an angular displacement of  $\theta = \pi/2$ . The phase shift  $\cos(\theta + \pi/2)$  becomes  $-\sin(\theta)$ . Poisson's equation becomes

$$u(r, \theta) = \frac{1}{2\pi} \frac{1 - r^2}{1 + r^2 + 2r \sin(\theta)}$$

#### Question: 8.1.16

With complex exponentials in  $F(x) = \sum c_k e^{ikx}$ , the energy identity (21) changes to  $\int_{-\pi}^{\pi} |F(x)|^2 dx = 2\pi \sum |c_k|^2$ . Derive this by integrating  $(\sum c_k e^{ikx})(\sum \bar{c}_k e^{-ikx})$ .

By orthogonality, we know that surviving terms will only be those with the same  $k$  in both summations. This leaves us with

$$\int_{-\pi}^{\pi} \sum |c_k|^2 dx = 2\pi \sum |c_k|^2$$

#### Question: 8.1.17

A centered square wave has  $F(x) = 1$  for  $|x| \leq \pi/2$ .

- Find its energy  $\int |F(x)|^2 dx$  by direct integration
- Compute its Fourier coefficients  $c_k$  as specific numbers
- Find the sum in the energy identity (Problem 16).

(a)

$$\int_{-\pi}^{\pi} |F(x)|^2 dx = \int_{-\pi/2}^{\pi/2} dx = \pi$$

(b)

$$\begin{aligned} c_k &= \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} e^{-ikx} dx \\ &= -\frac{1}{2\pi ik} (e^{-ik\pi/2} - e^{ik\pi/2}) \\ &= \frac{1}{\pi k} \sin\left(\frac{k\pi}{2}\right) \end{aligned}$$

In general, the squared coefficients take on form

$$|c_k|^2 = \begin{cases} \frac{1}{4}, & k = 0 \\ \frac{1}{\pi^2(2k-1)^2}, & k > 0 \end{cases}$$

- (c) Since we need to sum over all integers (positive, negative, and zero), we need to multiply the summation here by two to account for the negatives (as it only sums from  $k = 1$  onward), and add  $|c_0|^2$ .

$$\begin{aligned}
 \int_{-\pi}^{\pi} \sum |c_k|^2 dx &= 2\pi \sum_{-\infty}^{\infty} |c_k|^2 \\
 &= 2\pi \left( |c_0|^2 + \sum_{k=1}^{\infty} (|c_k|^2 + |c_{-k}|^2) \right) \\
 &= 2\pi \left( |c_0|^2 + 2 \sum_{k=1}^{\infty} |c_k|^2 \right) \\
 &= 2\pi \left( \frac{1}{4} + 2 \sum \frac{1}{\pi^2 (2k-1)^2} \right) \\
 &= 2\pi \left( \frac{1}{4} + \frac{1}{4} \right) \\
 &= \pi
 \end{aligned}$$

#### Question: 8.1.18

$F(x) = 1 + (\cos(x))/2 + \cdots + (\cos(nx))/2^n + \cdots$  is analytic: infinitely smooth.

- (a) If you take 10 derivatives, what is the Fourier series of  $d^{10}F/dx^{10}$ ?  
 (b) Does that series still converge quickly? Compare  $n^{10}$  with  $2^n$  for  $n = 2^{10}$ .

- (a) We have

$$F^{10}(x) = - \left[ \frac{\cos(x)}{2} + \cdots + \frac{n^{10} \cos(nx)}{2^n} + \cdots \right] = - \sum_{k=1}^{\infty} \frac{k^{10} \cos(kx)}{2^k}$$

which is indeed its own Fourier series.

- (b) Yes;  $(2^{10})^{10} = 2^{100}$  is immensely less than  $2^{2^{10}}$ . In general,  $n^{10}$  grows much slower than  $2^n$  for large  $n$ .

#### Question: 8.1.19

If  $f(x) = 1$  for  $|x| \leq \pi/2$  and  $f(x) = 0$  for  $\pi/2 < |x| < \pi$ , find its cosine coefficients. Can you graph and compute the Gibbs overshoot at the jumps?

The cosine coefficients are

$$a_k = \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} \cos(kx) dx = \frac{2}{\pi k} \sin\left(\frac{k\pi}{2}\right)$$

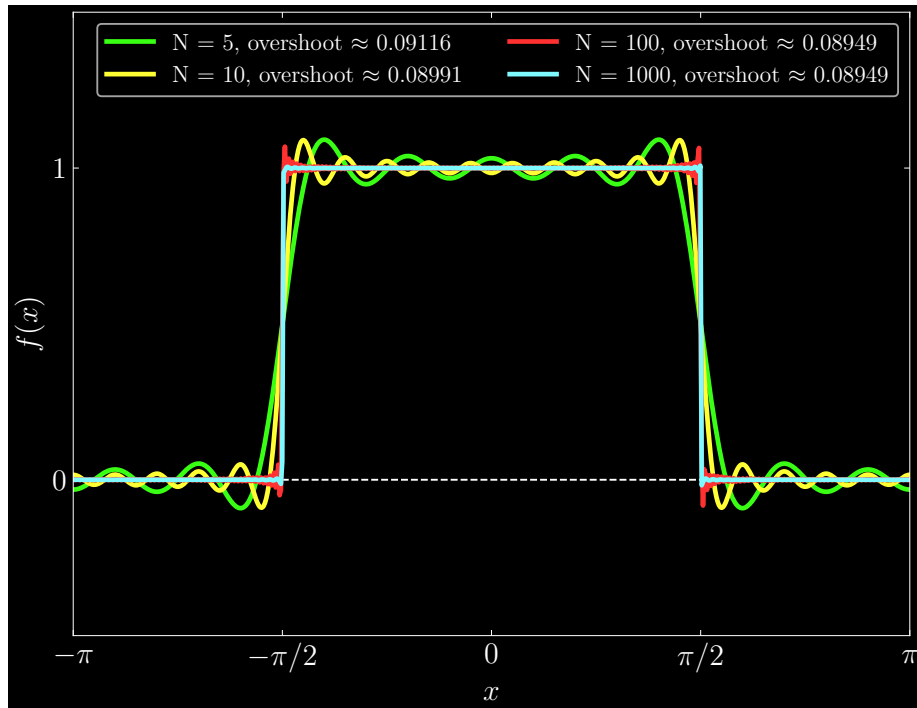
which reduces to

$$a_n = (-1)^{n+1} \frac{2}{\pi(2n-1)}$$

Since  $a_0 = 1/2$ , the Fourier cosine series is

$$f(x) = \frac{1}{2} + \sum_{n=1}^{\infty} (-1)^{n+1} \frac{2}{\pi(2n-1)} \cos((2n-1)x)$$

The Gibbs overshoot for numerous values of partial sums  $S_N$  is



**Question: 8.1.20**

Find all the coefficients  $a_k$  and  $b_k$  for  $F$ ,  $I$ , and  $D$  on the interval  $-\pi \leq x \leq \pi$ :

$$F(x) = \delta\left(x - \frac{\pi}{2}\right) \quad I(x) = \int_0^x \delta\left(x - \frac{\pi}{2}\right) dx \quad D(x) = \frac{d}{dx} \delta\left(x - \frac{\pi}{2}\right)$$

$F(x)$ :

$$\begin{aligned} a_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \delta\left(x - \frac{\pi}{2}\right) dx \\ a_k &= \frac{1}{\pi} \int_{-\pi}^{\pi} \cos(kx) \delta\left(x - \frac{\pi}{2}\right) dx \\ &= \frac{1}{\pi} \cos\left(\frac{k\pi}{2}\right) \\ b_k &= \frac{1}{\pi} \int_{-\pi}^{\pi} \sin(kx) \delta\left(x - \frac{\pi}{2}\right) dx \\ &= \frac{1}{\pi} \sin\left(\frac{k\pi}{2}\right) \end{aligned}$$

$I(x)$ : Note that this  $I(x)$  is a definition of the Heaviside step function  $H\left(x - \frac{\pi}{2}\right)$ .

$$\begin{aligned} a_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \int_0^x \delta\left(x - \frac{\pi}{2}\right) dx dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} H\left(x - \frac{\pi}{2}\right) dx \\ &= \frac{1}{4} \\ a_k &= \frac{1}{\pi} \int_{-\pi}^{\pi} \cos(kx) H\left(x - \frac{\pi}{2}\right) dx \\ &= \frac{1}{\pi k} \left[ \sin(k\pi) - \sin\left(\frac{k\pi}{2}\right) \right] \\ &= -\frac{1}{\pi k} \sin\left(\frac{k\pi}{2}\right) \\ b_k &= \frac{1}{\pi} \int_{-\pi}^{\pi} \sin(kx) H\left(x - \frac{\pi}{2}\right) dx \\ &= \frac{1}{\pi k} \left[ \cos\left(\frac{k\pi}{2}\right) - \cos(k\pi) \right] \end{aligned}$$

$D(x)$ : Relying on the intuition of integration by parts is key, although remember that the delta function is not a “function” in the rigorous sense (for reasons beyond our scope here).

For  $a_0$ , let  $u = 1$  and  $dv = \frac{d}{dx}\delta\left(x - \frac{\pi}{2}\right) dx$ . Then  $du = 0$  and  $v = \delta\left(x - \frac{\pi}{2}\right)$ .

$$\begin{aligned} a_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{d}{dx}\delta\left(x - \frac{\pi}{2}\right) dx \\ &= \frac{1}{2\pi} \delta\left(x - \frac{\pi}{2}\right) \Big|_{-\pi}^{\pi} \\ &= 0 \end{aligned}$$

In the cases of  $a_k$  and  $b_k$ , let  $u = \cos(kx)$  or  $u = \sin(kx)$ , corresponding to  $du = -k \sin(kx)$  or  $du = k \cos(kx)$ .

$$\begin{aligned} a_k &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos(kx) \frac{d}{dx}\delta\left(x - \frac{\pi}{2}\right) dx \\ &= -\frac{k}{2\pi} \int_{-\pi}^{\pi} \sin(kx) \delta\left(x - \frac{\pi}{2}\right) dx \\ &= -\frac{k}{2\pi} \sin\left(\frac{k\pi}{2}\right) \\ b_k &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \sin(kx) \frac{d}{dx}\delta\left(x - \frac{\pi}{2}\right) dx \\ &= \frac{k}{2\pi} \int_{-\pi}^{\pi} \cos(kx) \delta\left(x - \frac{\pi}{2}\right) dx \\ &= \frac{k}{2\pi} \cos(kx) \end{aligned}$$

Observe that  $a_k$  and  $b_k$  here are the Fourier coefficients of the step function case multiplied by  $k$ .

#### Question: 8.1.21

For the one-sided tall box function in Example 4, with  $F = 1/h$  for  $0 \leq x \leq h$ , what is its odd part  $\frac{1}{2}(F(x) - F(-x))$ ? I am surprised that the Fourier coefficients of this odd part disappear as  $h$  approaches zero and  $F(x)$  approaches  $\delta(x)$ .

In order for the even and odd decomposition to work, we should clarify that our  $F$  is defined over a symmetric interval  $-h \leq x \leq h$ , namely

$$F(x) = \begin{cases} 1/h, & 0 \leq x \leq h \\ 0, & -h \leq x < 0 \end{cases}$$

Given the piecewise nature of this function, we must calculate the odd part on each interval. First, over  $0 \leq x \leq h$ :

$$F_{\text{odd}} = \frac{1}{2} \left( \frac{1}{h} - 0 \right) = \frac{1}{2h}$$

and secondly, over  $-h \leq x < 0$ :

$$F_{\text{odd}} = \frac{1}{2} \left( 0 - \frac{1}{h} \right) = -\frac{1}{2h}$$

Now we calculate the Fourier coefficients

$$\begin{aligned}
 a_0 &= \frac{1}{2h} \int_{-h}^h F(x) \, dx \\
 &= \frac{1}{2h} \left[ \int_0^h \frac{1}{2h} \, dx - \int_{-h}^0 \frac{1}{2h} \, dx \right] \\
 &= \frac{1}{2h} \left[ \frac{1}{2} - \frac{1}{2} \right] \\
 &= 0 \\
 a_k &= \frac{1}{h} \int_{-h}^h F(x) \cos(kx) \, dx \\
 &= \frac{1}{2h^2} \left[ \int_0^h \cos(kx) \, dx - \int_{-h}^0 \cos(kx) \, dx \right] \\
 &= 0 \\
 b_k &= \frac{1}{\pi} \int_{-h}^h F(x) \sin(kx) \, dx \\
 &= \frac{1}{2\pi h} \left[ \int_0^h \sin(kx) \, dx - \int_{-h}^0 \sin(kx) \, dx \right] \\
 &= \frac{1}{2\pi kh} [-\cos(kh) + 1 + 1 - \cos(kh)] \\
 &= \frac{\cos(kh) - 1}{\pi kh}
 \end{aligned}$$

In the limit, we can apply l'Hôpital's rule to find that  $b_k \rightarrow 0$  as  $h \rightarrow 0$ , though  $b_k$  is never truly zero.

#### Question: 8.1.22

Find the series  $F(x) = \sum c_k e^{ikx}$  for  $F(x) = e^x$  on  $-\pi \leq x \leq \pi$ . That function  $e^x$  looks smooth, but there must be a hidden jump to get coefficients  $c_k$  proportional to  $1/k$ . Where is the jump?

The coefficients are

$$\begin{aligned}
 c_k &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^x e^{-ikx} \, dx \\
 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{(1-ik)x} \, dx \\
 &= \frac{1}{2\pi(1-ik)} e^{(1-ik)x} \Big|_{-\pi}^{\pi} \\
 &= \frac{1}{2\pi(1-ik)} [e^{(1-ik)\pi} - e^{-(1-ik)\pi}] \\
 &= \frac{(-1)^k}{2\pi(1-ik)} [e^{\pi} - e^{-\pi}]
 \end{aligned}$$

The jumps occur at every  $x = (2k + 1)\pi$ , where we go from  $e^\pi$  down to  $e^{-\pi}$ . That jump appears as a factor in the Fourier coefficients.

**Question: 8.1.23**

(a) (Old particular solution) Solve  $Ay'' + By' + Cy = e^{ikx}$ .

(b) (New particular solution) Solve  $Ay'' + By' + Cy = \sum c_k e^{ikx}$ .

(a) Let  $y_p = Y_k e^{ikx}$ . Then  $y'' = -k^2 Y_k e^{ikx}$  and  $y' = ik Y_k e^{ikx}$ . We have

$$-k^2 Y_k A + ik Y_k B + Y_k C = 1 \quad \implies \quad Y_k = \frac{1}{C - k^2 A + ikB}$$

(b) By superposition,

$$y_p = \sum c_k Y_k e^{ikx}$$



## 8.2 - The Fast Fourier Transform

### Question: 8.2.1

Multiply the three matrices in equation (11) and compare with  $F$ . In which six entries do you need to know that  $i^2 = -1$ ? This is  $(w_4)^2 = w_2$ . If  $M = N/2$ , why is  $(w_N^M = -1)$ ?

$$\begin{aligned}
 F_4 &= \begin{bmatrix} 1 & & 1 & \\ & 1 & & i \\ 1 & & -1 & \\ & 1 & & -i \end{bmatrix} \begin{bmatrix} 1 & 1 & & \\ & i^2 & & \\ & & 1 & 1 \\ & & 1 & i^2 \end{bmatrix} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i^2 & i & i^3 \\ 1 & 1 & -1 & -1 \\ 1 & i^2 & -i & -i^3 \end{bmatrix} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & i^2 & i^3 \\ 1 & -1 & 1 & -1 \\ 1 & -i & i^2 & -i^3 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & i^2 & i^3 \\ 1 & i^2 & i^4 & i^6 \\ 1 & i^3 & i^6 & i^9 \end{bmatrix}
 \end{aligned}$$

The right-most three entries of the bottom two rows require knowledge of  $i^2 = -1$ .

For  $M = N/2$ , we have

$$w_N^M = w_N^{N/2} = e^{i2\pi \cdot N/2N} = e^{i\pi} = -1$$

These are the roots of unity on opposite ends of the real axis.

### Question: 8.2.2

Why is row  $i$  of  $\overline{F}$  the same as row  $N - i$  of  $F$  (numbered from 0 to  $N - 1$ )?

I will switch the index to  $k$  as the book's choice is confusing. One intuitive, geometric analogy to make here is  $\overline{F}$  *reverses* the direction of the roots of unity starting from zero. As we move down the Fourier matrix row by row, instead of moving counterclockwise around the unit circle to the  $k$ -th step, the conjugate changes the sign, causing us to move in the clockwise direction.

The other argument is the powers of  $\omega = e^{i2\pi/N}$  are computed modulo  $N$ . The  $k$ -th row of  $\overline{F}$  has  $\omega = e^{-i2k\pi/N}$  and the  $N - k$ -th row of  $F$  has  $\omega = e^{i2\pi(N-k)/N} = e^{-i2k\pi/N}$ . In this sense, row " $N$ " of  $F$  simply corresponds to row 0: which is a row of 1's in both of  $F$  and  $\overline{F}$ .

**Question: 8.2.3**

From Problem 2, find the 4 by 4 permutation matrix  $P$  so that  $F = P\bar{F}$ . Check that  $P^2 = I$  so that  $P = P^{-1}$ . Then from  $\bar{F}F = 4I$  show that  $F^2 = 4P$ .

It is amazing that  $F^4 = 16P^2 = 16I$ . *Four transforms of any  $\mathbf{c}$  bring back  $\mathbf{16c}$ .* For all  $N$ ,  $F^2/N$  is a permutation matrix  $P$  and  $\mathbf{F}^4 = \mathbf{N}^2\mathbf{I}$ .

In the 4 by 4 case, we need only switch rows 1 and 3:

$$P = \begin{bmatrix} 1 & & & \\ & & 1 & \\ & 1 & & \\ & & & 1 \end{bmatrix}$$

Check that

$$P^2 = \begin{bmatrix} 1 & & & \\ & & 1 & \\ & 1 & & \\ & & & 1 \end{bmatrix} \begin{bmatrix} 1 & & & \\ & & 1 & \\ & 1 & & \\ & & & 1 \end{bmatrix} = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} = I$$

Next, we can prove

$$\begin{aligned} \bar{F}F &= PF^2 \quad (PF = \bar{F} \iff F = P\bar{F}) \\ \implies F^2 &= 4P \end{aligned}$$

**Question: 8.2.4**

Invert the three factors in equation (11) to find a fast factorization of  $F^{-1}$ .

$$\begin{aligned} \begin{bmatrix} 1 & & 1 & \\ & 1 & & i \\ 1 & & -1 & \\ & 1 & & -i \end{bmatrix}^{-1} &= \frac{1}{2} \begin{bmatrix} 1 & & 1 & \\ & 1 & & 1 \\ 1 & & -1 & \\ & -i & & i \end{bmatrix} \\ \begin{bmatrix} 1 & 1 & & \\ 1 & i^2 & & \\ & & 1 & 1 \\ & & 1 & i^2 \end{bmatrix}^{-1} &= \frac{1}{2} \begin{bmatrix} 1 & 1 & & \\ 1 & -1 & & \\ & & 1 & 1 \\ & & 1 & -1 \end{bmatrix} \\ \begin{bmatrix} 1 & & & \\ & & 1 & \\ & 1 & & \\ & & & 1 \end{bmatrix}^{-1} &= \begin{bmatrix} 1 & & & \\ & & 1 & \\ & 1 & & \\ & & & 1 \end{bmatrix} \end{aligned}$$

The fast factorization of  $F_4^{-1}$  is

$$F_4^{-1} = \frac{1}{4} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & & \\ 1 & -1 & & \\ & & 1 & 1 \\ & & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & & 1 & \\ & 1 & & 1 \\ 1 & & -1 & \\ & -i & & i \end{bmatrix}$$

**Question: 8.2.5**

$F$  is symmetric. Transpose equation (11) to find a new Fast Fourier Transform.

$$F^T = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & & \\ 1 & i^2 & & \\ & & 1 & 1 \\ & & 1 & i^2 \end{bmatrix} \begin{bmatrix} 1 & & 1 & \\ & 1 & & 1 \\ 1 & & -1 & \\ & i & & -i \end{bmatrix}$$

**Question: 8.2.6**

All entries in the factorization of  $F_6$  involve powers of  $w =$  sixth root of 1:

$$F_6 = \begin{bmatrix} I & D \\ I & -D \end{bmatrix} \begin{bmatrix} F_3 & \\ & F_3 \end{bmatrix} \begin{bmatrix} & P \\ & \end{bmatrix}$$

Write down these factors with 1,  $w$ ,  $w^2$  in  $D$  and powers of  $w^2$  in  $F_3$ . Multiply!

$$I = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & & \\ & w & \\ & & w^2 \end{bmatrix}, \quad F_3 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & w^2 & (w^2)^2 \\ 1 & (w^2)^2 & (w^2)^4 \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$$

A point of confusion is in conflating  $w_3$  and  $w_6$ , so to clarify, above we have  $w = w_6$ . Use the relationship  $(w_6)^2 = w_3$ . This gives us  $F_3$ . Multiplying the factors gives us

$$\begin{aligned}
F_6 &= \begin{bmatrix} 1 & & & & & \\ & 1 & & & & \\ & & 1 & & & \\ & & & -1 & & \\ & & & & -w & \\ & & & & & -w^2 \end{bmatrix} \begin{bmatrix} 1 & & & & & \\ & 1 & & & & \\ & & w^2 & & & \\ & & & (w^2)^2 & & \\ & & & & (w^2)^4 & \\ & & & & & 1 \end{bmatrix} \begin{bmatrix} 1 & & & & & \\ & & 1 & & & \\ & & & 1 & & \\ & & & & 1 & \\ & & & & & 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 & & & & & \\ & 1 & & & & \\ & & 1 & & & \\ & & & -1 & & \\ & & & & -w & \\ & & & & & -w^2 \end{bmatrix} \begin{bmatrix} 1 & & & & & \\ & 1 & & & & \\ & & w^2 & & & \\ & & & (w^2)^2 & & \\ & & & & (w^2)^4 & \\ & & & & & 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & w & w^2 & w^3 & (w^2)^2 & w^5 \\ 1 & w^2 & (w^2)^2 & (w^2)^3 & (w^2)^4 & (w^2)^5 \\ 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & -w & w^2 & -w^3 & (w^2)^2 & -w^5 \\ 1 & -w^2 & (w^2)^2 & -(w^2)^3 & (w^2)^2 & -(w^2)^5 \end{bmatrix}
\end{aligned}$$

**Question: 8.2.7**

Put the vector  $\mathbf{c} = (1, 0, 1, 0)$  through the three steps of the FFT to find  $\mathbf{y} = F\mathbf{c}$ . Do the same for  $\mathbf{c} = (0, 1, 0, 1)$ .

$$\begin{aligned}
F_4 c_1 &= \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -i \end{bmatrix} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & i^2 \end{bmatrix} \begin{bmatrix} 1 & & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \\
&= \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -i \end{bmatrix} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & i^2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -i \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} \\
&= \begin{bmatrix} 2 \\ 0 \\ 2 \\ 0 \end{bmatrix}
\end{aligned}$$

$$\begin{aligned}
F_4 c_2 &= \begin{bmatrix} 1 & 1 & & \\ & 1 & i & \\ 1 & & -1 & \\ & 1 & & -i \end{bmatrix} \begin{bmatrix} 1 & 1 & & \\ & 1 & i^2 & \\ & & 1 & 1 \\ & & 1 & i^2 \end{bmatrix} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 1 & & \\ & 1 & i & \\ 1 & & -1 & \\ & 1 & & -i \end{bmatrix} \begin{bmatrix} 1 & 1 & & \\ & 1 & i^2 & \\ & & 1 & 1 \\ & & 1 & i^2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 1 & & \\ & 1 & i & \\ 1 & & -1 & \\ & 1 & & -i \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix} \\
&= \begin{bmatrix} 2 \\ 0 \\ -2 \\ 0 \end{bmatrix}
\end{aligned}$$

**Question: 8.2.8**

Compute  $\mathbf{y} = F_8 \mathbf{c}$  by the three FFT steps for  $\mathbf{c} = (1, 0, 1, 0, 1, 0, 1, 0)$ . Repeat the computation for  $\mathbf{c} = (0, 1, 0, 1, 0, 1, 0, 1)$ .

Let  $w = w_8$ . We have

$$\begin{aligned}
F_8 &= \begin{bmatrix} I_4 & D_4 \\ I_4 & -D_4 \end{bmatrix} \begin{bmatrix} F_4 & \\ & F_4 \end{bmatrix} \begin{bmatrix} & P_8 \\ & \end{bmatrix} \\
I_4 &= \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}, \quad D_4 = \begin{bmatrix} 1 & & & \\ & w^2 & & \\ & & (w^2)^2 & \\ & & & (w^2)^4 \end{bmatrix}
\end{aligned}$$

where we use the fact that  $(w_8)^2 = w_4$ . The permutation matrix is

$$P_8 = \begin{bmatrix} 1 & & & & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & 1 & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & 1 & & & & \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & 1 & \end{bmatrix}$$

Through recursion, and further exploiting that  $(w_4)^2 = w_2 = w^4$ , we further decompose into

$$\begin{bmatrix} F_4 & \\ & F_4 \end{bmatrix} = \begin{bmatrix} I_2 & D_2 & & \\ I_2 & -D_2 & & \\ & & I_2 & D_2 \\ & & I_2 & -D_2 \end{bmatrix} \begin{bmatrix} F_2 & & & \\ & F_2 & & \\ & & F_2 & \\ & & & F_2 \end{bmatrix} \begin{bmatrix} \text{pick} & 0, 4 \\ \text{pick} & 2, 6 \\ \text{pick} & 1, 5 \\ \text{pick} & 3, 7 \end{bmatrix}$$

$$I_2 = \begin{bmatrix} 1 & \\ & 1 \end{bmatrix}, \quad D_2 = \begin{bmatrix} 1 & \\ & w^4 \end{bmatrix}, \quad F_2 = \begin{bmatrix} 1 & 1 \\ 1 & w^4 \end{bmatrix}$$

Computing,  $\mathbf{c}_1$  outputs  $(4, 0, 0, 0, 4, 0, 0, 0)$  and  $\mathbf{c}_2$  gives  $(4, 0, 0, 0, -4, 0, 0, 0)$ .

#### Question: 8.2.9

If  $w = e^{2\pi i/64}$  then  $w^2$  and  $\sqrt{w}$  are among the \_\_\_ and \_\_\_ roots of 1.

We have  $w^2$  is a primitive  $32^{th}$  root of unity and  $\sqrt{w}$  is a primitive  $128^{th}$  root of unity. Both generate the other roots of unity for their respective  $N$ .

#### Question: 8.2.10

$F$  is a symmetric matrix. Its eigenvalues aren't real. How is this possible?

The theorem requires that the symmetric matrix itself must be real to have real eigenvalues. By contrast,  $F$  definitionally has imaginary components in its entries, so that assumption is not satisfied.

#### Question: 8.2.11

The eigenvectors of  $K_N$  and  $B_N$  are the discrete sines  $\mathbf{s}_1, \dots, \mathbf{s}_N$  and the discrete cosines  $\mathbf{c}_0, \dots, \mathbf{c}_{N-1}$ . Notice the eigenvector  $\mathbf{c}_0 = (1, 1, \dots, 1)$ . Here are the  $\mathbf{s}_k$  and  $\mathbf{c}_k$ —these vectors are samples of  $\sin(kx)$  and  $\cos(kx)$  from 0 to  $\pi$ .

$$\left( \sin\left(\frac{\pi k}{N+1}\right), \sin\left(\frac{2\pi k}{N+1}\right), \dots, \sin\left(\frac{N\pi k}{N+1}\right) \right)$$

and

$$\left( \cos\left(\frac{\pi k}{2N}\right), \cos\left(\frac{3\pi k}{2N}\right), \dots, \cos\left(\frac{(2N-1)\pi k}{2N}\right) \right)$$

For 2 by 2 matrices  $K_2$  and  $B_2$ , verify that  $\mathbf{s}_1, \mathbf{s}_2$  and  $\mathbf{c}_0, \mathbf{c}_1$  are eigenvectors.

See the inset on the symmetric tridiagonal matrices in the textbook. In the cases of  $K_2$  and  $B_2$ , we have

$$K_2 = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \quad B_2 = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

The eigenvectors are

$$\mathbf{s}_1 = \begin{bmatrix} \sqrt{3}/2 \\ \sqrt{3}/2 \end{bmatrix}, \quad \mathbf{s}_2 = \begin{bmatrix} \sqrt{3}/2 \\ -\sqrt{3}/2 \end{bmatrix}$$

$$\mathbf{c}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{c}_1 = \begin{bmatrix} \sqrt{2}/2 \\ -\sqrt{2}/2 \end{bmatrix}$$

Multiplying, we find

$$\begin{aligned} K_2 \mathbf{s}_1 &= \mathbf{s}_1, & K_2 \mathbf{s}_2 &= 3\mathbf{s}_2 \\ B_2 \mathbf{c}_0 &= \mathbf{0}, & B_2 \mathbf{c}_1 &= 2\mathbf{c}_1 \end{aligned}$$

#### Question: 8.2.12

Show that  $C_3$  has eigenvalues  $\lambda = 0, 3, 3$  with eigenvectors  $\mathbf{e}_0 = (1, 1, 1)$ ,  $\mathbf{e}_1 = (1, w, w^2)$ ,  $\mathbf{e}_2 = (1, w^2, w^4)$ . You may prefer the real eigenvectors  $(1, 1, 1)$  and  $(1, 0, -1)$  and  $(1, -2, 1)$ .

We have

$$C_3 = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$

Now, let  $w = w_3$  be a third root of unity. This is easiest if we rely on our geometric intuition. Envision the roots of unity as complex vectors tracing out the unit circle. Treat each equation as a vector addition problem. Doing so allows us to easily calculate

$$\begin{aligned} C_3 \mathbf{e}_0 &= \mathbf{0} \\ C_3 \mathbf{e}_1 &= \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ w \\ w^2 \end{bmatrix} = \begin{bmatrix} 2 - w - w^2 \\ -1 + 2w - w^2 \\ -1 - w + 2w^2 \end{bmatrix} = 3\mathbf{e}_1 \\ C_3 \mathbf{e}_2 &= \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ w^2 \\ w^4 \end{bmatrix} = \begin{bmatrix} 2 - w^2 - w^4 \\ -1 + 2w^2 - w^4 \\ -1 - w^2 + 2w^4 \end{bmatrix} = 3\mathbf{e}_2 \end{aligned}$$

#### Question: 8.2.13

Multiply to see the eigenvectors  $\mathbf{e}_k$  and eigenvalues  $\lambda_k$  of  $C_N$ . Simplify to  $\lambda_k = 2 - 2\cos(2\pi k/N)$ . Explain why  $C_N$  is only semidefinite. It is not positive definite.

$$C\mathbf{e}_k = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ w^k \\ w^{2k} \\ w^{(N-1)k} \end{bmatrix} = (2 - w^k - w^{-k}) \begin{bmatrix} 1 \\ w^k \\ w^{2k} \\ w^{(N-1)k} \end{bmatrix}$$

We can derive

$$\lambda_k = 2 - w^k - w^{-k} = 2 - (e^{i2\pi k/N} + e^{-i2\pi k/N}) = 2 - 2\cos(2\pi k/N)$$

Immediately, we can conclude that  $\lambda_k \geq 0$  for all  $k$ , as the cosine term is bounded by 1. We have  $\lambda_k = 0$  only when  $k$  is a multiple of  $N$ . Equivalently,  $C_N$  is semidefinite.

**Question: 8.2.14**

The eigenvectors  $\mathbf{e}_k$  of  $C$  are automatically perpendicular because  $C$  is a      matrix. (To tell the truth,  $C$  has repeated eigenvalues as in Problem 12. There was a plane of eigenvectors for  $\lambda = 3$  and we chose orthogonal  $\mathbf{e}_1$  and  $\mathbf{e}_2$  in that plane.)

$C$  is a **symmetric** matrix, which always has orthogonal eigenvectors.

**Question: 8.2.15**

Write the 2 eigenvalues for  $K_2$  and the 3 eigenvalues for  $B_3$ . Always  $K_N$  and  $B_{N+1}$  have the same  $N$  eigenvalues, with the extra eigenvalue      for  $B_{N+1}$ . (This is because  $K = A^T A$  and  $B = A A^T$ .)

The characteristic polynomials are

$$\begin{aligned}
 K_2 &= \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \\
 \implies (2 - \lambda)^2 - 1 &= 0 \\
 \implies \lambda_{1,2} &= 1, 3 \\
 B_3 &= \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \\
 \implies (1 - \lambda)[(2 - \lambda)(1 - \lambda) - 1] - (1 - \lambda) &= 0 \\
 = (1 - \lambda)[(2 - \lambda)(1 - \lambda) - 2] &= 0 \\
 \implies \lambda_{1,2,3} &= 0, 1, 3
 \end{aligned}$$

The extra eigenvalue of  $B_3$  is  $\lambda = 0$ .



### 8.3 - The Heat Equation

#### Question: 8.3.1

Solve the heat equation  $u_t = cu_{xx}$  on an infinite bar with coefficient  $c$ , starting from  $u = e^{ikx}$  at  $t = 0$ . As in (10) the solution has the product form  $u = e^{i\omega t} e^{ikx}$ . With  $c$  in the equation, find  $\omega$  for each  $k$ .

With the given solution form, write

$$u_t = (i\omega) e^{i\omega t} e^{ikx} = c(ik)^2 e^{i\omega t} e^{ikx} = u_{xx}$$

Setting the coefficients equal gives us

$$i\omega = c(ik)^2 \implies i\omega = -ck^2$$

#### Question: 8.3.2

Solve the same equation  $u_t = cu_{xx}$  starting from the point source  $u = \delta(x) = \int e^{ikx} dk / 2\pi$  at  $t = 0$ . By superposition, you integrate over all  $k$  the solutions  $u$  in Problem 1. The result is the heat kernel as in equation (14) but adjusted for  $c$ .

By superposition, the solution with the given initial condition is

$$U(t, x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ck^2 t} e^{ikx} dk$$

To solve, differentiate by  $x$ :

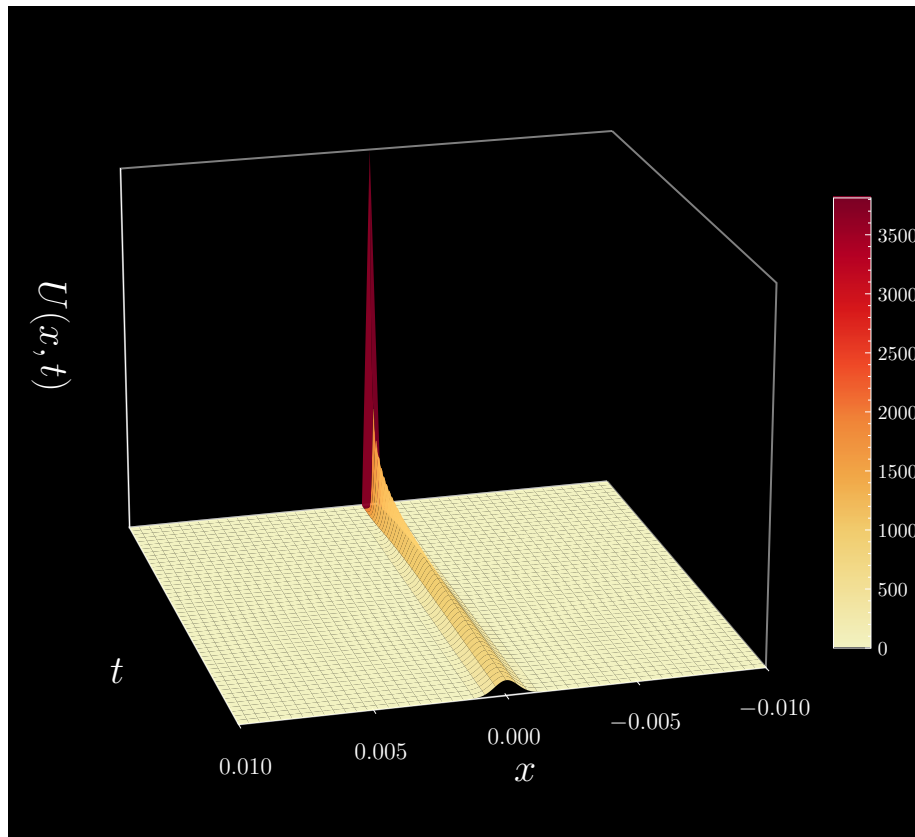
$$\frac{d}{dx} U(t, x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ck^2 t} (ik) e^{ikx} dk$$

Proceed by integration by parts. Let  $a = ie^{ikx}$  and  $db = ke^{-ck^2 t} dk$ . Then  $da = -xe^{ikx} dk$  and  $b = -(1/2ct) e^{-ck^2 t}$ , and we have

$$\begin{aligned} \frac{dU}{dx} &= \frac{1}{2\pi} \left( -\frac{i}{2} cte^{ikx} e^{-ck^2 t} \Big|_{-\infty}^{\infty} - \frac{x}{2ct} \int_{-\infty}^{\infty} e^{-ck^2 t} e^{ikx} dk \right) \\ &= -\frac{x}{2ct} U \\ \implies \frac{dU}{U} &= -\frac{x}{2ct} dx \\ \implies U &= Me^{-x^2/4ct} \end{aligned}$$

Assume a heat kernel of  $\int \delta(x) dx = 1$ . To normalize this Gaussian integral, set  $M = 1/\sqrt{4\pi ct}$ . Then our solution is

$$U = \frac{1}{\sqrt{4\pi ct}} e^{-x^2/4ct}$$



### Question: 8.3.3

To solve  $u_t = cu_{xx}$  for a bar between  $x = 0$  and  $x = 1$ , the basis functions are still  $\sin(k\pi x)$  (with  $u = 0$  at the ends). What are the eigenvalues  $\lambda_k$  that go into the solution  $\sum b_k(0) e^{-\lambda_k t} \sin(k\pi x)$ ?

With the diffusion constant, we now have

$$\lambda_k = ck^2\pi^2$$

On the left,  $\partial u / \partial t$  gives us  $ck^2\pi^2$ . On the right,  $\partial^2 u / \partial x^2$  gives us  $k^2\pi^2$ . Multiply by the diffusivity constant for equality.

**Question: 8.3.4**

Following Problem 3, solve  $u_t = cu_{xx}$  when the initial temperature is  $u_0 = 1$  for  $\frac{1}{4} \leq x \leq \frac{3}{4}$  (and  $u_0 = 0$  on the first and last quarters of the bar). The problem is to find the coefficients  $b_k(0)$  for that initial temperature.

First, derive the Fourier series for the even square wave

$$SW(x) = \begin{cases} 1 & \text{for } -\frac{3}{4} < x < -\frac{1}{4} \\ 0 & \text{for } -1 < x < -\frac{3}{4} \text{ and } -\frac{1}{4} < x < \frac{1}{4} \text{ and } \frac{3}{4} < x < 1 \\ 1 & \text{for } \frac{1}{4} < x < \frac{3}{4} \end{cases}$$

An intuitive approach here is to alter the Fourier series for the square wave given in the book's equation (6):

$$\frac{4}{\pi} \sum_{k=1}^{\infty} \frac{\sin[(2k-1)\pi x]}{2k-1}$$

First is to offset the horizontal position by  $3/4$  and adjust the periodicity by a factor of 2: instead of the width of the period being 1, it is now  $1/2$ .

$$SW(x) = \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{\sin[(4k-2)\pi(x + \frac{3}{4})]}{2k-1}$$

By the angle difference identity for sine and reducing trigonometric functions to even-odd sign parity, this becomes

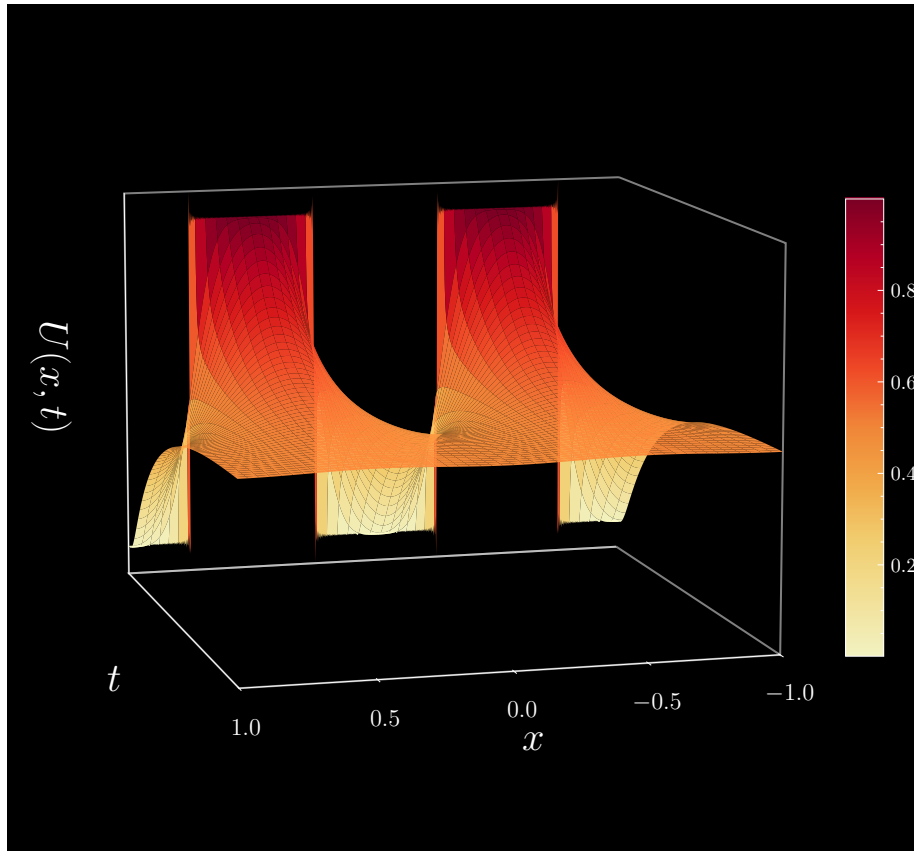
$$SW(x) = \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{(-1)^k}{2k-1} \cos[(4k-2)\pi x]$$

Lastly, we need to scale the amplitude and vertically offset the square wave. For this problem, multiply the sum by  $1/2$  (since the wave peaks between 0 and 1) and add  $1/2$  (since it is symmetric about  $u = 1/2$ ). This gives us our transformed square wave.

$$SW(x) = \frac{1}{2} + \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{(-1)^k}{2k-1} \cos[(4k-2)\pi x]$$

The corresponding coefficients are then

$$b_k(0) = \frac{2}{\pi} \frac{(-1)^k}{2k-1}$$

**Question: 8.3.5**

Solve the heat equation  $u_t = u_{xx}$  from a point source  $u(x, 0) = \delta(x)$  with free boundary conditions  $u'(\pi, t) = u'(-\pi, t) = 0$ . Use the infinite cosine series  $\delta(x) = (1 + 2 \cos(x) + 2 \cos(2x) + \dots) / 2\pi$  multiplied by time decay factors  $b_k(t)$ .

Incorporating the time decay factors, the solution is

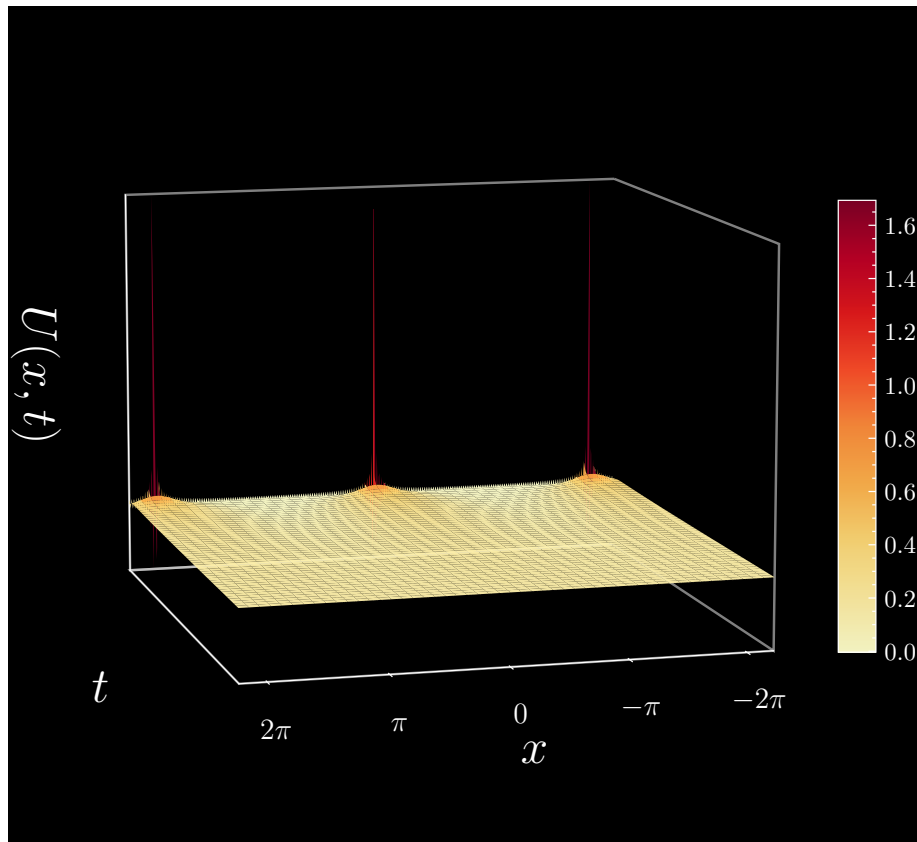
$$u(x, t) = \frac{1}{2\pi} + \frac{1}{\pi} \sum_{k=1}^{\infty} e^{-k^2 \pi^2 t} \cos(kx)$$

Note that the Fourier coefficients to the cosines series provide us with the  $b_k(0)$  constants. Differentiating with respect to  $x$ , we get

$$u'(x, t) = -\frac{1}{\pi} \sum_{k=1}^{\infty} e^{-k^2 \pi^2 t} k \sin(kx)$$

From which it is immediately apparent that

$$u'(\pi, t) = u'(-\pi, t) = 0$$

**Question: 8.3.6**

(Bar from  $x = 0$  to  $x = \infty$ ) Solve  $u_t = u_{xx}$  on the positive half of an infinite bar, starting from the shifted delta function  $u_0 = \delta(x - a)$  at a point  $x = a > 0$ . Here is a way to use the full-bar heat kernel  $U$  in (14), and still keep  $u = 0$  at  $x = 0$ .

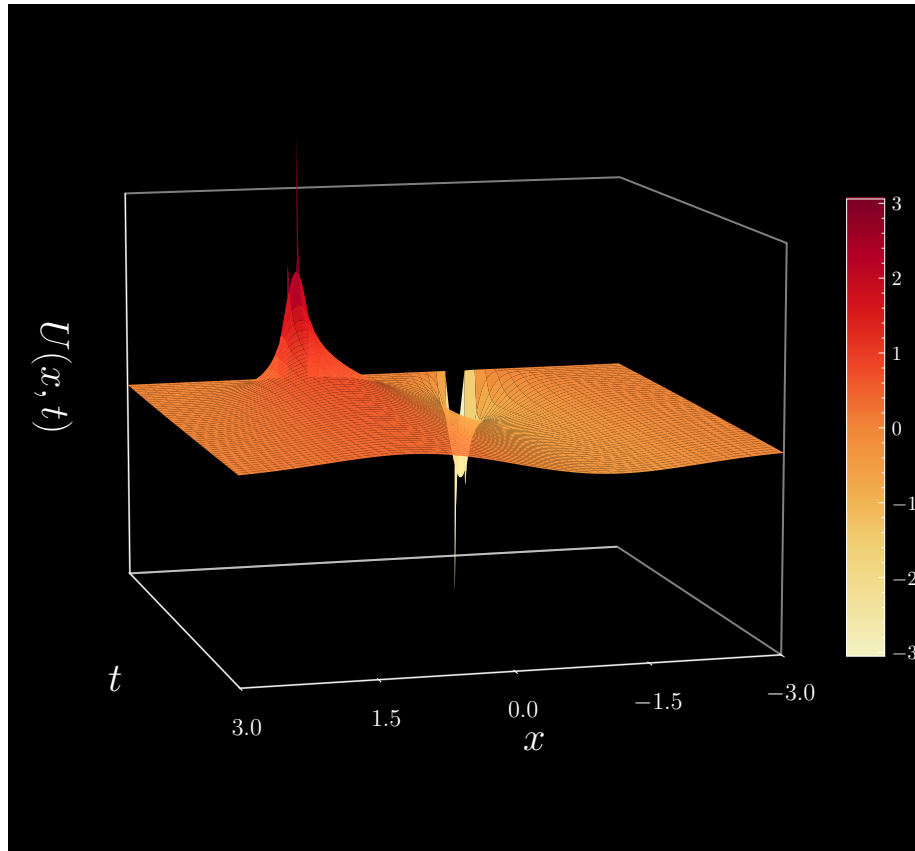
Imagine a negative point source at  $x = -a$ . Solve the heat equation on the fully infinite bar, including both sources in  $u_0 = \delta(x - a) - \delta(x + a)$  at  $t = 0$ . Your solution (a difference of heat kernels) will stay zero at the boundary  $x = 0$  (*Why?*). Then it must be the correct solution on the half-bar, since it started correctly.

Using equation (14) and invoking superposition, our solution is

$$u(t, x) = \frac{1}{\sqrt{4\pi t}} \left[ e^{-(x-a)^2/4t} - e^{-(x+a)^2/4t} \right]$$

Note that at  $x = 0$ ,  $u(t, 0) = 0$ . This method of solution is called the **method of images**. It is called so because of the trick in dropping in another source that generates the desired boundary condition.

For  $a = 1$ , we can graph the solution as

**Question: 8.3.7**

Check that the basis functions  $s_k = \sin\left(k + \frac{1}{2}\right) \pi x$  are orthogonal over  $0 \leq x \leq 1$ . Find a formula for the coefficient  $B_4$  in the Fourier series  $F(x) = \sum B_k s_k$ . (Multiply by  $s_4(x)$  and integrate, to isolate  $B_4$ .)

Let  $k_1 \neq k_2$  be integers. Then we can ascertain

$$\begin{aligned} \int_0^1 \sin\left(k_1 + \frac{1}{2}\right) \pi x \sin\left(k_2 + \frac{1}{2}\right) \pi x \, dx &= \frac{1}{2} \int_0^1 [\cos(k_1 - k_2) \pi x - \cos(k_1 + k_2 + 1) \pi x] \, dx \\ &= \frac{1}{2\pi} \left[ \frac{\sin(k_1 - k_2) \pi}{k_1 - k_2} - \frac{\sin(k_1 + k_2 + 1) \pi}{k_1 + k_2 + 1} \right] \\ &= 0 \end{aligned}$$

This is true for any choice of  $k_1, k_2$ . Invoking orthogonality, we can then calcu-

late

$$\begin{aligned}\int_0^1 F(x) s_4(x) dx &= \int_0^1 \sum B_k s_k s_4 dx \\ &= B_4 \int_0^1 \sin^2\left(\frac{9\pi x}{2}\right) dx \\ &= \frac{B_4}{2}\end{aligned}$$

Leaving us with

$$B_4 = 2 \int_0^1 F(x) s_4(x) dx$$

#### Question: 8.3.8

The basis functions  $\sin(k + \frac{1}{2})\pi x$  are for **fixed-free** boundaries ( $u = 0$  at  $x = 0$  and  $u' = 0$  at  $x = 1$ ). What are the basis functions for **free-fixed boundaries** ( $u' = 0$  at  $x = 0$  and  $u = 0$  at  $x = 1$ )?

For free-fixed boundaries, our basis functions are  $\cos(k + \frac{1}{2})\pi x$ . Easy to verify that the free-fixed boundary conditions are met.

#### Question: 8.3.9

Suppose  $\mathbf{u}_t = \mathbf{u}_{xx} - \mathbf{u}$  with boundary condition  $u = 0$  at  $x = 0$  and  $x = 1$ . Find the new numbers  $\lambda_k$  in the general solution  $u = \sum b_k(0) e^{-\lambda_k t} \sin(k\pi x)$ . (Previously  $\lambda_k = -k^2\pi^2$ , now there is a new term in  $\lambda$  because of  $-u$ .)

The additional  $-u$  term adds one to the eigenvalue after canceling the negative, giving us  $\lambda_k = k^2\pi^2 + 1$ .

#### Question: 8.3.10

Explain each step in equation (13). Solve  $dU/dx = -xU/2t$  to reach  $U = e^{-x^2/4t}$ . How do the known infinite integrals  $\int e^{-x^2} dx = \sqrt{\pi}$  and  $\int u dx = 1$  lead to the factor  $1/\sqrt{4\pi t}$ ?

The integration by parts proceeds as follows: let  $a = ie^{ikx}$  and  $db = e^{-k^2 t} k dk$ . Then  $da = -xe^{ikx} dk$  and  $b = -e^{-k^2 t}/2t$ . Combining the parts (don't forget the  $1/2\pi$  term) gives

$$\begin{aligned}\frac{dU}{dx} &= -ie^{-k^2 t} e^{ikx} \Big|_{-\infty}^{\infty} - \frac{1}{4\pi t} \int_{-\infty}^{\infty} (e^{-k^2 t}) (xe^{ikx}) dk \\ &= -\frac{1}{4\pi t} \int_{-\infty}^{\infty} (e^{-k^2 t}) (xe^{ikx}) dk \\ &= -\frac{xU}{2t}\end{aligned}$$

Separating variables, we can deduce from  $dU/dx = -xU/2t$  that

$$\int \frac{dU}{U} = - \int \frac{x}{2t} dx \implies \ln(U) = -\frac{x^2}{2t} + C \implies U = ce^{-x^2/2t}$$

The normalization constant  $c$  follows from the requirement that

$$\int_{-\infty}^{\infty} ce^{-x^2/2t} dx = 1$$

which is true only when  $c = 1/\sqrt{4\pi t}$ .

#### Question: 8.3.11

**(Shift invariance)** What is the solution to  $u_t = u_{xx}$  starting from  $\delta(x - a)$  at  $t = 0$ ?

Equation (18): an integral of shifted heat kernels.

$$u(t, x) = \frac{1}{\sqrt{4\pi t}} \int_{-\infty}^{\infty} u_0(a) e^{-(x-a)^2/4t} da$$

#### Question: 8.3.12

What are basis functions  $A(x, y)$  for heat flow in a square plate, when  $u = 0$  along the four sides  $x = 0$ ,  $x = 1$ ,  $y = 0$ ,  $y = 1$ ? The heat equation is  $u_t = u_{xx} + u_{yy}$ . Find eigenfunctions for  $A_{xx} + A_{yy} = \lambda A$  that satisfy the boundary conditions.

The first eigenfunction is  $A_{11} = \sin(\pi x) \sin(\pi y)$ . Find the eigenvalues  $\lambda$ .

For heat flowing out of the perimeter of the square, we have basis functions

$$A(x, y) = \sin(m\pi x) \sin(n\pi y)$$

The eigenfunctions and corresponding eigenvalues are

$$\begin{aligned} -\pi^2(m^2 + n^2) \sin(m\pi x) \sin(n\pi y) &= \lambda_{m,n} \sin(m\pi x) \sin(n\pi y) \\ \implies \lambda_{m,n} &= -\pi^2(m^2 + n^2) \end{aligned}$$

#### Question: 8.3.13

Substitute  $U = e^{-x^2/4t}/\sqrt{4\pi t}$  to show that this heat kernel solves  $U_t = U_{xx}$ .

Calculate

$$U_t = \left[ \frac{x^2}{4t^2} - \frac{1}{2t} \right] \frac{e^{-x^2/4t}}{\sqrt{4\pi t}} = U_{xx}$$



## 8.4 - The Wave Equation

Problems 1-4 are about the one-way wave equation  $\partial u / \partial t = c \partial u / \partial x$ .

### Question: 8.4.1

Suppose  $u(0, x) = \sin(2x)$ . What is the solution to  $u_t = cu_x$ ? At which times  $t_1, t_2, \dots$  will the solution return to the initial condition  $\sin(2x)$ ?

The solution is  $u(t, x) = \sin(2(x + ct))$ . The initial condition is returned to for all times  $t_k = k\pi/c$ .

### Question: 8.4.2

Suppose  $u_0(x) = \delta(x)$ , a big bang at the origin of the one-dimensional universe. At time  $t$  the bang is heard at the point  $x = \_\_\_\_$ . For  $u_{tt} = c^2 u_{xx}$  the bang will reach the two points  $x = \_\_\_\_$  and  $x = \_\_\_\_$  at time  $t$ .

In the one-directional case, the solution is  $u(t, x) = \delta(x + ct)$  and the bang will be heard at  $x = -ct$ . For two directions, we have

$$u(t, x) = \frac{1}{2} [\delta(x - ct) + \delta(x + ct)]$$

in which the bang is heard at  $x = ct$  and  $x = -ct$  at time  $t$ .

### Question: 8.4.3

- (a) Integrate both sides of  $u_t = cu_x$  from  $x = -\infty$  to  $\infty$  to prove that the total mass  $M = \int u \, dx$  is constant:  $dM/dt = 0$ .
- (b) Multiply by  $u$  and integrate both sides of  $uu_t = cuu_x$  to prove that  $E = \int u^2 \, dx$  is constant.

- (a) The problem should specify that  $u \rightarrow 0$  as  $|x| \rightarrow \infty$ . Intuitively, we should also conceive of  $u$  as a “density” of sorts. Combined, these insights suggest a spatial localization of the source of the wave (for instance, if I perturb the water of the Atlantic Ocean off of the coast of Coney Island, New York, it is completely unlikely that someone at Normandy beach in France will know about it). Derive

$$\begin{aligned} \int_{-\infty}^{\infty} u_t \, dx &= \frac{d}{dt} \int_{-\infty}^{\infty} u \, dx \\ \int_{-\infty}^{\infty} cu_x \, dx &= \int_{-\infty}^{\infty} cu'_0(x + ct) \, dx \\ &= c[u_0(\infty + ct) - u_0(-\infty + ct)] \\ &= 0 \end{aligned}$$

Equating the two sides establishes conservation of mass.

(b) Now we derive conservation of energy.

$$\begin{aligned}\int_{-\infty}^{\infty} uu_t dx &= \frac{1}{2} \frac{d}{dt} \int_{-\infty}^{\infty} u^2 dx \\ \int_{-\infty}^{\infty} cuu_x dx &= \frac{c}{2} \int_{-\infty}^{\infty} \frac{d}{dx} u^2 dx \\ &= 0\end{aligned}$$

#### Question: 8.4.4

Is the wave  $u(t, x) = u_0(x + ct)$  traveling left or right if  $c > 0$ ? To solve  $u_t = cu_x$  on the halfline  $0 \leq x < \infty$ , why is a boundary condition  $u(t, 0) = 0$  *not wanted*? With  $c < 0$  and waves in the opposite direction, that condition is appropriate.

For positive  $c$ , the wave is traveling left. The given boundary condition is problematic as it forces  $u = 0$  for all  $t$  at  $x = 0$ . In other words, whichever initial conditions the solution has, its profile at the outflow must be zero. With negative  $c$ , the wave travels right from the inflow point at  $x = 0$ , and the boundary condition simply requires that  $u$  begins at zero.

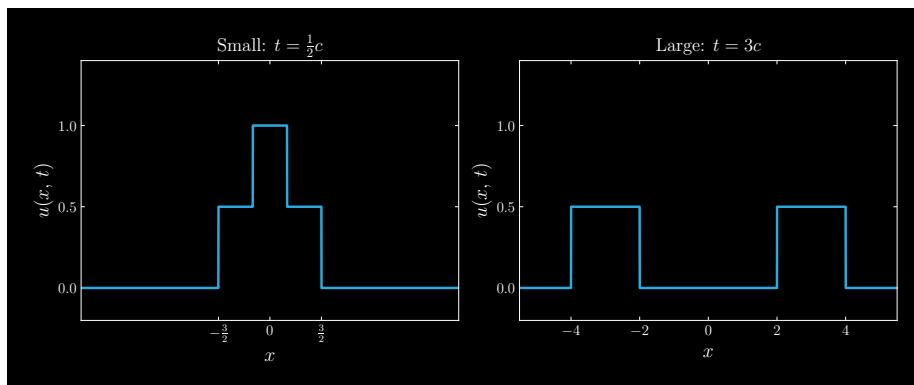
**Problems 5-9 are about the one-dimensional wave equation**

$$\partial^2 u / \partial t^2 = c^2 \partial^2 u / \partial x^2$$

#### Question: 8.4.5

A “box of water” has  $u_0(x) = 1$  for  $-1 \leq x \leq 1$ . Starting with zero velocity  $v_0(x)$ , the wave equation  $u_{tt} = c^2 u_{xx}$  is solved by  $u(t, x) = \frac{1}{2}u_0(x + ct) + \frac{1}{2}u_0(x - ct)$ . Graph this solution for small  $t = \frac{1}{2}c$  and large  $t = 3c$ .

For simplicity, set  $c = 1$ . The intuition is clear: the box of water “separates” into two waves, propagating anti-parallel from the center.



**Question: 8.4.6**

Under a flat ocean with  $u_0(x) = 1$ , an earthquake produces  $v_0(x) = \delta(x)$ . A one-dimensional tsunami starts moving with speed  $c$ . What is the solution (7) at time  $t$ ?

In equation (7), the last term becomes

$$\frac{1}{2c} \int_{x-ct}^{x+ct} \delta(x) dx = \begin{cases} 0, & 0 \notin [x-ct, x+ct] \\ 1, & 0 \in [x-ct, x+ct] \end{cases}$$

And since the ocean is flat, the first two terms simply sum to unity. This all corresponds to solution

$$u(t, x) = 1 + \begin{cases} \frac{1}{2c}, & |x| \leq ct \\ 0 & \text{otherwise} \end{cases}$$

**Question: 8.4.7**

Separation of variables gives  $u(t, x) = \sin(nx) \sin(nct)$  and three other similar solutions to  $u_{tt} = c^2 u_{xx}$ . *What are those three?* Which complex functions  $e^{ikx} e^{i\omega t}$  solve the wave equation?

We also have

$$\begin{aligned} u(t, x) &= \cos(nx) \cos(nct) \\ u(t, x) &= \sin(nx) \cos(nct) \\ u(t, x) &= \cos(nx) \sin(nct) \end{aligned}$$

From superposition, we may deduce that complex functions of the form

$$u(t, x) = e^{inx} e^{inct}$$

comprehensively encapsulate the four trigonometric solutions to the wave equation.

**Question: 8.4.8**

The 3D wave equation  $u_{tt} = u_{xx} + u_{yy} + u_{zz}$  becomes 1D when  $u$  has spherical symmetry:  $u$  depends only on  $r$  and  $t$ .

$$r = \sqrt{x^2 + y^2 + z^2} \quad \text{and} \quad \frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial r^2} + \frac{2}{r} \frac{\partial u}{\partial r}$$

- Multiply by  $r$  to find  $(ru)_{tt} = (ru)_{rr}$ !* Then  $ru$  is a function of  $r+t$  and  $r-t$ .
- Describe the solution  $ru = \delta(r-t-1)$ . This spherical sound wave has the radius  $r = \underline{\hspace{1cm}}$  at  $t = 8$ .

(a) Derive

$$\begin{aligned}
 (ru)_{tt} &= r_{tt}u + ru_{tt} \\
 &= ru_{tt} \\
 (ru)_r &= u + ru_r \\
 (ru)_{rr} &= u_r + (u_r + ru_{rr}) \\
 &= 2u_r + ru_{rr}
 \end{aligned}$$

(b) We can analogize this to the front of an “explosion” with initial radius 1. As time progresses to  $t = 8$ , the explosion front grows to  $r = 9$ .

#### Question: 8.4.9

The wave equation along a bar with density  $\rho$  and stiffness  $k$  is  $(\rho u_t)_t = (k u_x)_x$ . What is the velocity  $c$  in  $u_{tt} = c^2 u_{xx}$ ? What is  $\omega$  in  $u = \sin(\pi x/L) \cos(\omega t)$ ?

Assuming constancy in the density and stiffness, we have  $c = \sqrt{k/\rho}$ . It follows that

$$\omega = \frac{\pi}{L} \sqrt{\frac{k}{\rho}}$$

#### Question: 8.4.10

The small vibrations of a beam satisfy the fourth order equation  $u_{tt} = -c^2 u_{xxxx}$ . Look for solutions  $u = X(x)T(t)$  and find separate equations for the functions  $X$  and  $T$ . Then find four solutions  $X(x)$  when  $T(t) = \cos(\omega t)$ .

For our setup, we have in this equation

$$X(x)T''(t) = -c^2 X''''(x)T(t)$$

Divide through by  $c^2 X(x)T(t)$  and set equal to  $-\omega^2$  to get

$$\frac{T''(t)}{c^2 T(t)} = -\frac{X''''(x)}{X(x)} = -\omega^2$$

Then to solve the temporal and spatial component respectively, we satisfy

$$T''(t) = -c^2 \omega^2 T(t) \quad X''''(x) = \omega^2 X(x)$$

Then  $T(t) = \sin(c\omega t)$  or  $\cos(c\omega t)$ , and  $X(x) = \sin(\sqrt{\omega}x)$ ,  $\cos(\sqrt{\omega}x)$ ,  $e^{\sqrt{\omega}x}$ , or  $e^{-\sqrt{\omega}x}$ .

When  $T(t) = \cos(\omega t)$ , solutions for  $X(x)$  include  $\sin(ax)$ ,  $\cos(ax)$ ,  $e^{ax}$ , and  $e^{-ax}$ , where  $a = \sqrt{\omega/c}$ .

The key difference here is whether you divide through by  $c^2$  in deriving the temporal and spatial equations, which will determine whether  $c$  shows up in one or the other.

**Question: 8.4.11**

If that beam is clamped ( $u = 0$  and  $\partial u / \partial x = 0$  at both ends  $x = 0$  and  $x = L$ ), show that the frequencies  $\omega$  in Problem 10 must have  $\cos(\omega L) \cosh(\omega L) = 1$ .

In setting up the solution, we will rewrite the basis functions  $e^{ax}$  and  $e^{-ax}$  in terms of  $\cosh(ax)$  and  $\sinh(ax)$ . The general set of solutions of the spatial component and its derivative will be of form

$$\begin{aligned} X(x) &= a_1 \sin(\omega x) + a_2 \cos(\omega x) + a_3 \sinh(\omega x) + a_4 \cosh(\omega x) \\ X'(x) &= \omega[a_1 \cos(\omega x) - a_2 \sin(\omega x) + a_3 \cosh(\omega x) + a_4 \sinh(\omega x)] \end{aligned}$$

We do not wish to impose any constraints on the coefficients beyond those that stem from the boundary conditions; remember this fact for later.

Now, impose the boundary conditions at  $x = 0$ .

$$\begin{aligned} X(0) &= a_2 + a_4 = 0 \\ X'(0) &= \omega(a_1 + a_3) = 0 \end{aligned}$$

Immediately, we reduce four unknowns to two:  $a_4 = -a_2$  and  $a_3 = -a_1$ . Taking this into account while incorporating the boundary conditions at  $x = L$ , we derive the system of equations

$$\begin{aligned} X(L) &= a_1 \underbrace{\left[ \sin(\omega L) - \sinh(\omega L) \right]}_p + a_2 \underbrace{\left[ \cos(\omega L) - \cosh(\omega L) \right]}_q = 0 \\ X'(L) &= a_1 \underbrace{\left[ \cos(\omega L) - \cosh(\omega L) \right]}_q - a_2 \underbrace{\left[ \sin(\omega L) + \sinh(\omega L) \right]}_r = 0 \end{aligned}$$

where we divide the second equation through by  $\omega$  as we won't need that. Rewriting this as a matrix equation, we have

$$\begin{bmatrix} p & q \\ q & -r \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Now, recall that we have to account for a *general* set of solutions. This means that there cannot be a unique solution for this system of equations. Specifically, we must have infinitely many solutions (coefficients). Therefore, we can immediately conclude that we must have *zero determinant*. Calculating, we can come to our final deduction:

$$\begin{aligned} \det &= -pr - q^2 \\ &= -[\sin(\omega L) - \sinh(\omega L)][\sin(\omega L) + \sinh(\omega L)] - [\cos(\omega L) - \cosh(\omega L)]^2 \\ &= -[\sin^2(\omega L) + \cos^2(\omega L)] - [\cosh^2(\omega L) - \sinh^2(\omega L)] + 2 \cos(\omega L) \cosh(\omega L) \\ &= -2 + 2 \cos(\omega L) \cosh(\omega L) \\ &= 0 \end{aligned}$$

which forces  $\cos(\omega L) \cosh(\omega L) = 1$ .

**Problems 12-16 solve the wave equation with boundary conditions at  $x = 0$  and  $x = L$ .**

**Question: 8.4.12**

A string plucked halfway along has  $u_0(x) = \delta(x - \frac{L}{2})$  and  $v_0(x) = 0$ . Find the Fourier coefficients  $b_k$  from equation (18). Write the first three terms of the Fourier series solution in (16).

The Fourier coefficients are

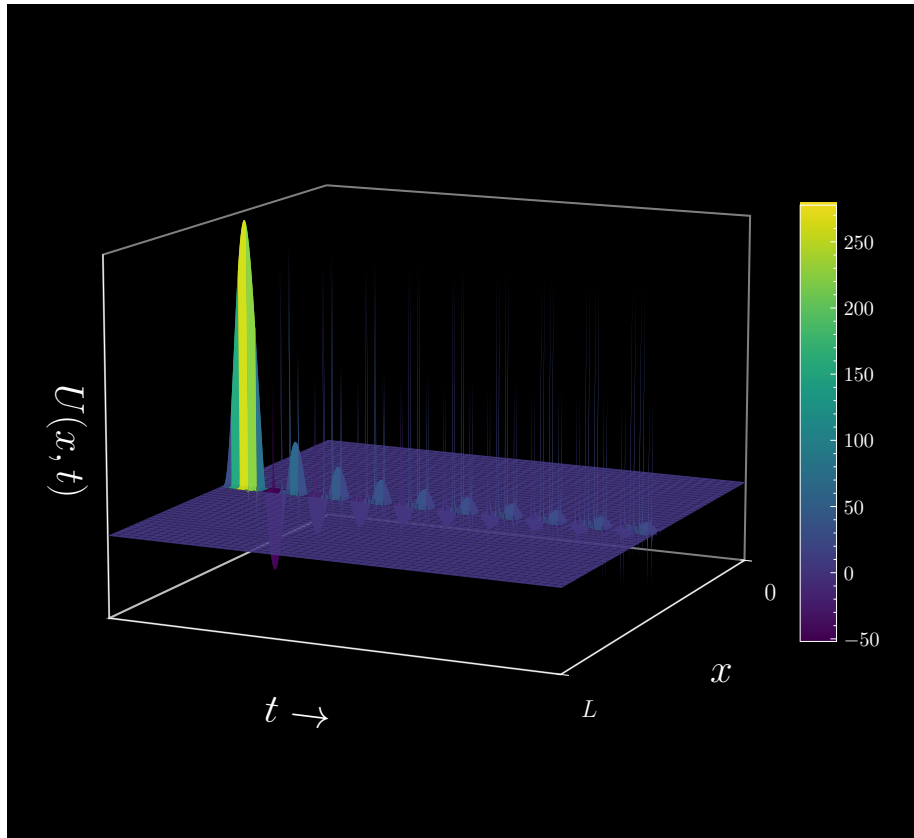
$$b_k = \frac{\int_0^L \delta\left(x - \frac{L}{2}\right) \sin\left(\frac{k\pi x}{L}\right) dx}{\int_0^L \sin^2\left(\frac{k\pi x}{L}\right) dx} = \frac{4k\pi \sin\left(\frac{k\pi}{2}\right)}{2kL\pi - L\sin(2k\pi)} = \frac{2}{L} \sin\left(\frac{k\pi}{2}\right)$$

We see that  $b_k$  vanishes for even  $k$ . Therefore, the full Fourier series is

$$u(t, x) = \sum_{k=1}^{\infty} \frac{2}{L} (-1)^{k+1} \sin\left(\frac{(2k-1)\pi x}{L}\right) \cos\left(\frac{(2k-1)\pi ct}{L}\right)$$

And its first three terms are

$$u(t, x) \approx \frac{2}{L} \left[ \sin\left(\frac{\pi x}{L}\right) \cos\left(\frac{\pi ct}{L}\right) - \sin\left(\frac{3\pi x}{L}\right) \cos\left(\frac{3\pi ct}{L}\right) + \sin\left(\frac{5\pi x}{L}\right) \cos\left(\frac{5\pi ct}{L}\right) \right]$$



**Question: 8.4.13**

Suppose the string starts with zero velocity  $v_0(x)$  from a *hat function*:  $u_0(x) = 2x/L$  for  $x < L/2$  and  $u_0(x) = 2(L-x)/L$  for  $x > L/2$ . Find the Fourier coefficients  $b_k$  from (18) and the first two nonzero terms of  $u(t, x)$  in (16).

Calculate

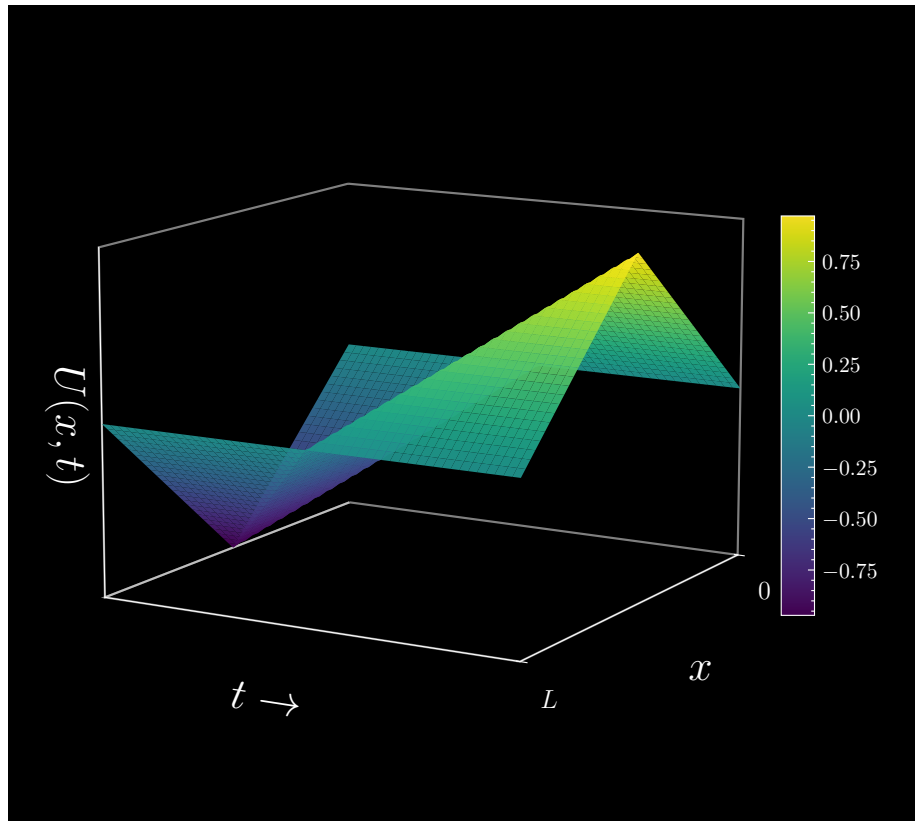
$$\frac{2}{L} \left[ \int_0^{L/2} x \sin\left(\frac{k\pi x}{L}\right) dx + \int_{L/2}^L (L-x) \sin\left(\frac{k\pi x}{L}\right) dx \right] = \frac{4L}{k^2\pi^2} \sin\left(\frac{k\pi}{2}\right)$$

Divide through by  $\int_0^L \sin^2(k\pi x/L) dx = L/2$  to get as our Fourier coefficients

$$b_k = \frac{8}{k^2\pi^2} \sin\left(\frac{k\pi}{2}\right)$$

The coefficients vanish for even  $k$ . Let  $k = 2n - 1$ . Then the solution is

$$u(t, x) = \sum_{n=1}^{\infty} \frac{8}{(2n-1)^2\pi^2} (-1)^{n-1} \sin\left(\frac{(2n-1)\pi x}{L}\right) \cos\left(\frac{(2n-1)\pi ct}{L}\right)$$

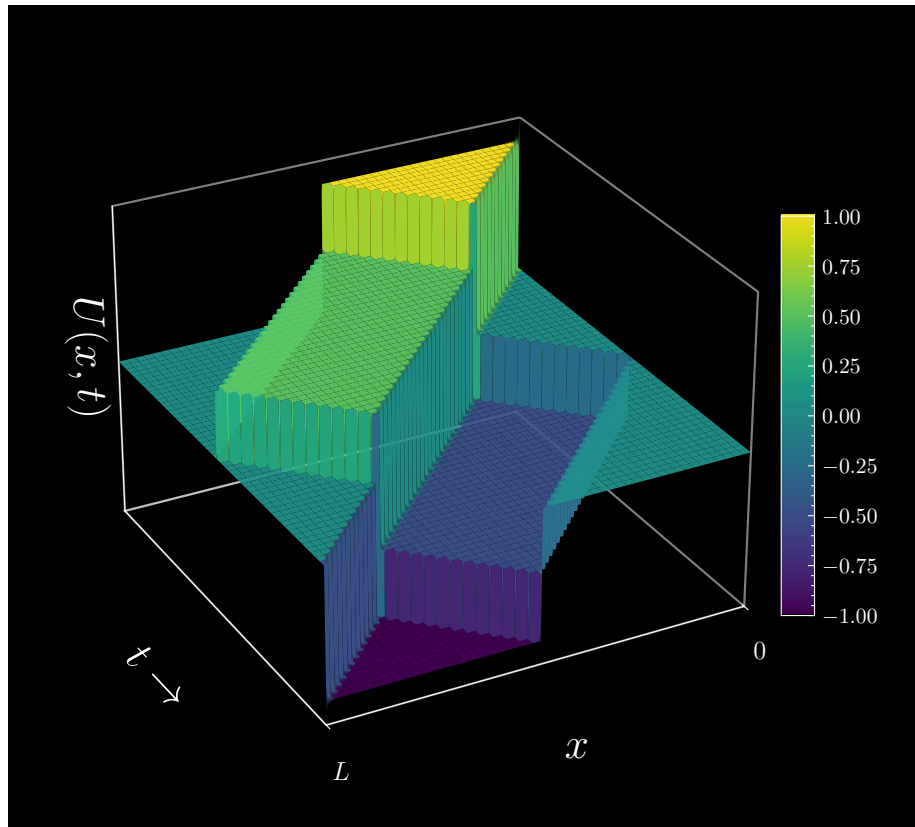


**Question: 8.4.14**

Suppose the string starts with zero velocity  $v_0(x)$  from a *box function*:  $u_0(x) = 1$  for  $x < L/2$ . Find all the  $b_k$  in the solution  $u = \sum b_k \sin(k\pi x/L) \cos(k\pi ct/L)$ .

Calculate the Fourier coefficients

$$b_k = \frac{\int_0^{L/2} \sin\left(\frac{k\pi x}{L}\right) dx}{\int_0^L \sin^2\left(\frac{k\pi x}{L}\right) dx} = \frac{2}{k\pi} \left[ 1 - \cos\left(\frac{k\pi}{2}\right) \right]$$

**Question: 8.4.15**

The boundary condition at a *free end*  $x = L$  is  $\partial u/\partial x = 0$  instead of  $u = 0$ . Solve  $X'' + \omega^2 X = 0$  to find  $X(x)$  and all allowable  $\omega$ 's with this new condition. Then solve  $T'' + \omega^2 c^2 T = 0$  to complete the solution  $u = \sum a_n X(x) T(t)$ .

Two null solutions:  $\sin(\omega x)$  and  $\cos(\omega x)$ . Imposing the boundary conditions:  $u = 0$  at  $x = 0$  and  $\partial u/\partial x = 0$  at  $x = L$ , we must have  $X(x) = \sin(\omega x)$  where



$\omega L = \pi/2 + n\pi$ . We can then deduce the values of  $\omega_n$  are

$$\omega_n = \frac{\pi(2n+1)}{2L}$$

For  $T(t)$ , once again we have two null solutions:  $\sin(\omega ct)$  and  $\cos(\omega ct)$ . Assuming initial condition of  $T' = v_0 = 0$ , only the cosine term will survive. Our full solution is then

$$u = \sum_{n=0}^{\infty} a_n \sin(\omega_n x) \cos(\omega_n ct)$$

with  $\omega_n$  defined as above. Note the summation beginning at  $n = 0$  in this case, so we start from  $\omega_0 = \pi/2L$ .

#### Question: 8.4.16

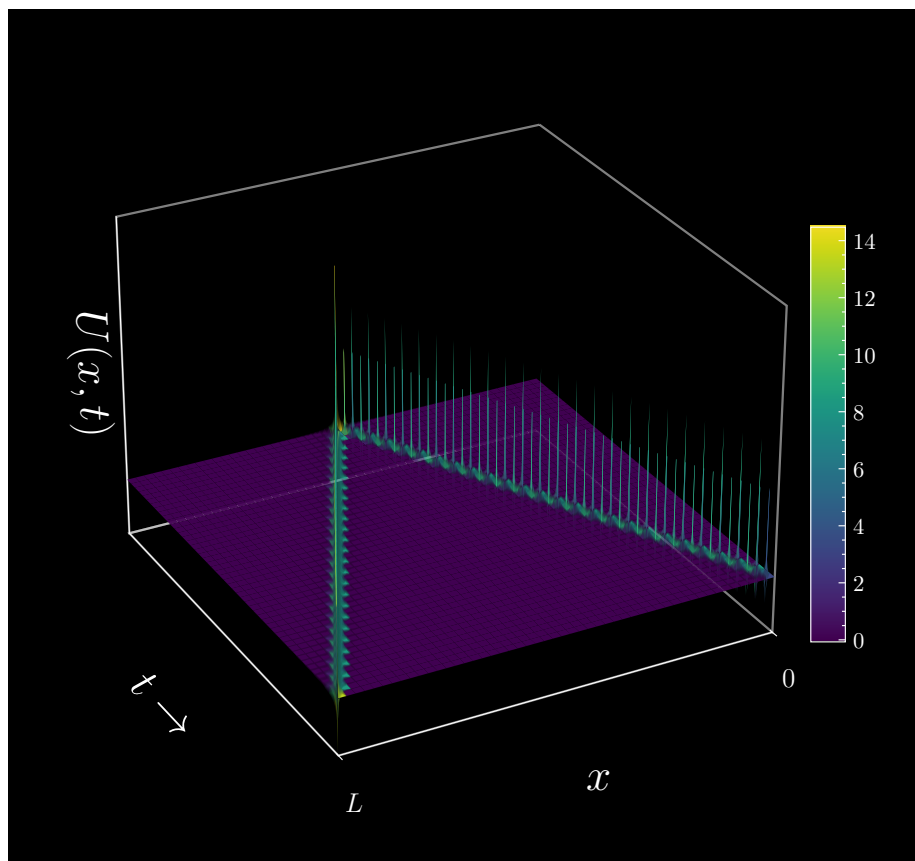
What is the solution  $u(t, x)$  on a string of length  $L = 2$  if  $u(0, x) = \delta(x - 1)$ ? The end  $x = 0$  is fixed by  $u(t, 0) = 0$  and the end  $x = 2$  is free:  $\partial u / \partial x(t, 2) = 0$ .

Using the solution form of Problem 15, we calculate the Fourier coefficients as (note the denominator here is simply 1 with  $L = 2$ )

$$b_n = \int_0^2 \delta(x - 1) \sin(\omega_n x) dx = \sin(\omega_n) = \sin\left(\frac{\pi(2n+1)}{4}\right)$$

With the boundary conditions, the solution is

$$u(t, x) = \sum_{n=0}^{\infty} \sin\left(\frac{\pi(2n+1)}{4}\right) \sin\left(\frac{\pi(2n+1)}{4} x\right) \cos\left(\frac{\pi(2n+1)}{4} t\right)$$



## 8.5 - The Laplace Transform

### Question: 8.5.1

When the driving function is  $f(t) = \delta(t)$ , the solution starting from rest is the **impulse response**. The impulse is  $\delta(t)$ , the response is  $y(t)$ . Transform this equation to find the **transfer function**  $Y(s)$ . Invert to find the impulse response  $y(t)$ .

$$y'' + y = \delta(t) \text{ with } y(0) = 0 \text{ and } y'(0) = 0$$

Since  $\mathcal{L}[y'] = sY(s) - y(0)$ , it follows that

$$\mathcal{L}[y''] = s\mathcal{L}[y'] - y'(0) = s^2\mathcal{L}[y] - sy(0) - y'(0)$$

The Laplace transform of the delta function can be derived as

$$\mathcal{L}[\delta(t)] = \int_0^\infty \delta(t) e^{-st} dt = 1$$

Then we have

$$s^2Y(s) + Y(s) = 1 \implies Y(s) = \frac{1}{s^2 + 1}$$

Inverting back, we can conclude

$$y(t) = \sin(t)$$

One confusion that might arise here is why  $y'(0) = \cos(0) = 1$  appears to be inconsistent with the initial condition. Recall that  $y(t)$  and its derivative are defined as zero for  $t < 0$ . Thus for  $y'(t)$ , we have a jump at  $t = 0$  from 0 to 1. I will get a little hand-wavey here, but imagine an instant before the impulse response kicks in, we do indeed start from an initial condition of rest, and then once the impulse hits, that derivative jumps to 1.

### Question: 8.5.2

(Important) Find the first derivative and second derivative of  $f(t) = \sin(t)$  for  $t \geq 0$ . Watch for a jump at  $t = 0$  which produces a spike (delta function) in the derivative.

We have

$$f'(t) = \begin{cases} 0 & t < 0 \\ \cos(t) & t \geq 0 \end{cases}$$

$$f''(t) = \begin{cases} 0 & t < 0 \\ -\sin(t) + \delta(t) & t \geq 0 \end{cases}$$

As we alluded to in the previous exercise, there is indeed a jump in  $f'(t)$  as we cross the threshold over  $t = 0$ , hence the  $\delta(t)$  term in the second derivative.

**Question: 8.5.3**

Find the Laplace transform of the unit box function

$$b(t) = \{1 \text{ for } 0 \leq t < 1\} = H(t) - H(t - 1)$$

The unit step function is  $H(t)$  in honor of Oliver Heaviside.

The Laplace transform of the Heaviside function  $H(t - T)$  is  $e^{-sT}/s$ . By linearity of the Laplace transform, we have

$$\mathcal{L}[b(t)] = \frac{1}{s} - \frac{e^{-s}}{s}$$

**Question: 8.5.4**

If the Fourier transform of  $f(t)$  is defined by  $\hat{f}(k) = \int f(t) e^{-ikt} dt$  and  $f(t) = 0$  for  $t < 0$ , what is the connection between  $\hat{f}(k)$  and the Laplace transform  $F(s)$ ?

When  $k = s/i$ ,  $\hat{f}(k) = F(s)$ .

**Question: 8.5.5**

What is the Laplace transform  $R(s)$  of the standard **ramp function**  $r(t) = t$ ? For  $t < 0$  all functions are zero. The derivative of  $r(t)$  is the unit step  $H(t)$ . Then multiplying  $R(s)$  by  $s$  gives \_\_\_\_.

From the transform for  $t^n e^{ct}$ , we have  $\mathcal{L}[r(t)] = 1/s^2$ . Multiplying by  $s$  gives  $1/s$ , for which the inverse transform takes us to 1 on  $t \geq 0$ : the derivative of  $r(t) = t$ , or the step function.

**Question: 8.5.6**

Find the Laplace transform  $F(s)$  of each  $f(t)$ , and the poles of  $F(s)$ :

- (a)  $f = 1 + t$
- (b)  $f = t \cos(\omega t)$
- (c)  $f = \cos(\omega t - \theta)$
- (d)  $f = \cos^2(t)$
- (e)  $f = e^{-2t} \cos(t)$
- (f)  $f = te^{-t} \sin(\omega t)$

Let us attempt to derive these transforms without the use of integrals; that is, refer to Strang's Table of Transforms and attempt to build on these results.

(a)  $F(s) = \frac{1}{s} + \frac{1}{s^2} = \frac{1+s}{s^2}$ , double pole  $s = 0$

- (b) Consider  $g(t) = te^{i\omega t}$ . Its Laplace transform is

$$\mathcal{L}[g(t)] = \frac{1}{(s - i\omega)^2} = \frac{1}{s^2 - \omega^2 - 2i\omega s} = \frac{s^2 - \omega^2 + 2i\omega s}{(s^2 - \omega^2)^2 + 4\omega^2 s^2} = \frac{s^2 - \omega^2 + 2i\omega s}{(s^2 + \omega^2)^2}$$

Take the real part, which indeed is the desired transform.

$$\mathcal{L}[f(t)] = \frac{s^2 - \omega^2}{(s^2 + \omega^2)^2}$$

which has poles  $s = \pm i\omega$ .

- (c) Use the cosine angle difference identity:

$$\cos(\omega t - \theta) = \cos(\omega t) \cos(\theta) + \sin(\omega t) \sin(\theta)$$

Then we can conclude

$$\mathcal{L}[f(t)] = \cos(\theta) \frac{s}{s^2 + \omega^2} + \sin(\theta) \frac{\omega}{s^2 + \omega^2}$$

which has poles  $s = \pm i\omega$ .

- (d) From the double-angle cosine identity, we have

$$\cos^2(t) = \frac{1 + \cos(2t)}{2}$$

which corresponds to transform

$$\mathcal{L}[f(t)] = \frac{1}{2s} + \frac{1}{2} \frac{s}{s^2 + 4} = \frac{s^2 + 2}{s(s^2 + 4)}$$

and has poles  $s = 0$  and  $s = \pm 2i$ .

- (e) Consider  $g(t) = e^{(-2+i)t}$ , which has transform

$$\mathcal{L}[g(t)] = \frac{1}{s + 2 - i} = \frac{s + 2 + i}{(s + 2)^2 + 1}$$

The real part corresponds to our desired transform.

$$\mathcal{L}[f(t)] = \frac{s + 2}{(s + 2)^2 + 1}$$

which has poles  $s = -2 \pm i$ .

- (f) Consider  $g(t) = te^{(-1+i\omega)t}$ , which has transform

$$\begin{aligned} \mathcal{L}[g(t)] &= \frac{1}{(s + 1 - i\omega)^2} = \frac{1}{(s + 1)^2 - \omega^2 - 2i\omega(s + 1)} \\ &= \frac{(s + 1)^2 - \omega^2 + 2i\omega(s + 1)}{\left((s + 1)^2 - \omega^2\right)^2 + 4\omega^2(s + 1)^2} \\ &= \frac{(s + 1)^2 - \omega^2 + 2i\omega(s + 1)}{\left((s + 1)^2 + \omega^2\right)^2} \end{aligned}$$

Whittling this down to the real part, the desired transform is

$$\mathcal{L}[f(t)] = \frac{(s+1)^2 - \omega^2}{((s+1)^2 + \omega^2)^2}$$

which has poles  $s = -1 \pm i\omega$ .

#### Question: 8.5.7

Find the Laplace transform  $s$  of  $f(t) = \text{next integer above } t$  and  $f(t) = t\delta(t)$ .

Fundamentally, the ceiling function is a staircase: a bunch of **step functions** joined together. It has a summation expression, namely

$$f(t) = \sum_{k=0}^{\infty} H(t-k)$$

By linearity, the Laplace transform is

$$F(s) = \sum_{k=0}^{\infty} \frac{e^{-sk}}{s} = \frac{1}{s(1-e^{-s})}$$

As for the second function, observe that it is zero everywhere. Therefore, its transform is also zero.

#### Question: 8.5.8

*Inverse Laplace Transform:* Find the function  $f(t)$  from its transform  $F(s)$ :

(a)  $\frac{1}{s-2\pi i}$

(b)  $\frac{s+1}{s^2+1}$

(c)  $\frac{1}{(s-1)(s-2)}$

(d)  $1/(s^2+2s+10)$

(e)  $e^{-s}/(s-a)$

(f)  $2s$

(a)  $f(t) = e^{2\pi i t}$

(b)  $f(t) = \cos(t) + \sin(t)$

(c) By partial fractions, we have

$$F(s) = \frac{1}{(s-1)(s-2)} = \frac{1}{s-2} - \frac{1}{s-1}$$

which corresponds to  $f(t) = e^{2t} - e^t$ .

(d) Decompose with partial fractions to get

$$F(s) = \frac{1}{s^2 + 2s + 10} = \frac{1}{(s+1-3i)(s+1+3i)} = \frac{1}{6i} \left( \frac{1}{s+1-3i} - \frac{1}{s+1+3i} \right)$$

and derive corresponding function

$$\begin{aligned} f(t) &= \frac{1}{6i} \left( e^{(-1+3i)t} - e^{(-1-3i)t} \right) \\ &= \frac{1}{3} e^{-t} \sin(3t) \end{aligned}$$

(e) The numerator indicates a temporal shift by  $T = 1$ . The function begins at  $t = 1$ , and we can express this as  $f(t) = e^{a(t-1)} H(t-1)$ . Verify by solving

$$\int_0^\infty e^{a(t-1)} H(t-1) e^{-st} dt = \int_1^\infty e^{a(t-1)} e^{-st} dt = \frac{e^{-s}}{s-a}$$

(f) Recall that for any  $f(t)$ , the transform of its derivative is

$$\mathcal{L}[f'(t)] = sF(s) - f(0)$$

If we can find an  $f(t)$  such that its transform is 1, then its derivative will be our solution. We previously derived that  $\mathcal{L}[\delta(t)] = 1$ , which implies that  $\delta'(t)$  is our target function. Indeed, it is true that

$$\int_0^\infty \delta'(t) e^{-st} dt = -(-se^{-st})_{(t=0)} = s$$

You may be wondering how this is consistent with the presence of the  $-f(0)$  term, when clearly  $\delta(0) \neq 0$ . The reason for this is subtle and beyond the scope of this textbook, but one point to consider is the delta function is a misnomer: it is technically *not a function in the traditional sense*. Rather, it is what we call a **generalized function**. This obviates the need for the  $f(0)$  condition.

#### Question: 8.5.9

Solve  $y'' + y = 0$  from  $y(0)$  and  $y'(0)$  by expressing  $Y(s)$  as a combination of  $s/(s^2 + 1)$  and  $1/(s^2 + 1)$ . Find the inverse transform  $y(t)$  from the table.

In the  $s$ -domain, this equation is

$$(s^2 + 1) \mathcal{L}[y] - sy(0) - y'(0) = 0$$

Isolating  $\mathcal{L}[y]$  gives us

$$\mathcal{L}[y] = \frac{s}{s^2 + 1} y(0) + \frac{1}{s^2 + 1} y'(0)$$

Transforming back to the  $t$ -domain, we have

$$y(t) = \cos(t) y(0) + \sin(t) y'(0)$$

**Question: 8.5.10**

Solve  $y'' + 3y' + 2y = \delta$  starting from  $y(0) = 0$  and  $y'(0) = 1$  by Laplace transform. Find the poles and partial fractions for  $Y(s)$  and invert to find  $y(t)$ .

Transforming to  $s$ -domain, we find

$$s^2 \mathcal{L}[y] - 1 + 3s \mathcal{L}[y] + 2 \mathcal{L}[y] = (s^2 + 3s + 2) \mathcal{L}[y] - 1 = 1$$

Isolating  $\mathcal{L}[y]$  gives us

$$\mathcal{L}[y] = \frac{2}{s^2 + 3s + 2} = \frac{2}{(s+2)(s+1)} = -\frac{2}{s+2} + \frac{2}{s+1}$$

In the  $t$ -domain, this is

$$y(t) = -2e^{-2t} + 2e^{-t}$$



**Question: 8.5.11**

Solve these initial-value problems by Laplace transform:

- (a)  $y' + y = e^{i\omega t}$ ,  $y(0) = 8$
- (b)  $y'' - y = e^t$ ,  $y(0) = 0$ ,  $y'(0) = 0$
- (c)  $y' + y = e^{-t}$ ,  $y(0) = 2$
- (d)  $y'' + y = 6t$ ,  $y(0) = 0$ ,  $y'(0) = 0$
- (e)  $y' - i\omega y = \delta(t)$ ,  $y(0) = 0$
- (f)  $my'' + cy' + ky = 0$ ,  $y(0) = 1$ ,  $y'(0) = 0$

(a)

$$\begin{aligned}
 s\mathcal{L}[y] - 8 + \mathcal{L}[y] &= (s+1)\mathcal{L}[y] - 8 \\
 &= \frac{1}{s-i\omega} \\
 \Rightarrow \mathcal{L}[y] &= \frac{1}{(s+1)(s-i\omega)} + \frac{8}{s+1} \\
 &= \frac{1}{1+i\omega} \left( -\frac{1}{s+1} + \frac{1}{s-i\omega} \right) + \frac{8}{s+1} \\
 \Rightarrow y(t) &= \frac{1}{1+i\omega} (-e^{-t} + e^{i\omega t}) + 8e^{-t}
 \end{aligned}$$

(b)

$$\begin{aligned}
 s^2\mathcal{L}[y] - \mathcal{L}[y] &= (s^2-1)\mathcal{L}[y] \\
 &= \frac{1}{s-1} \\
 \Rightarrow \mathcal{L}[y] &= \frac{1}{(s^2-1)(s-1)} \\
 &= \frac{1}{4(s+1)} - \frac{1}{4(s-1)} + \frac{1}{2(s-1)^2} \\
 \Rightarrow y(t) &= \frac{1}{4}e^{-t} - \frac{1}{4}e^t + \frac{1}{2}te^t
 \end{aligned}$$

(c)

$$\begin{aligned}
 s\mathcal{L}[y] - 2 + \mathcal{L}[y] &= (s+1)\mathcal{L}[y] - 2 \\
 &= \frac{1}{s+1} \\
 \Rightarrow \mathcal{L}[y] &= \frac{1}{(s+1)^2} + \frac{2}{s+1} \\
 \Rightarrow y(t) &= te^{-t} + 2e^{-t}
 \end{aligned}$$

(d)

$$\begin{aligned}
s^2 \mathcal{L}[y] + \mathcal{L}[y] &= (s^2 + 1) \mathcal{L}[y] \\
&= \frac{6}{s^2} \\
\Rightarrow \mathcal{L}[y] &= \frac{6}{s^2(s^2 + 1)} \\
&= \frac{6}{s^2} + \frac{3i}{s - i} - \frac{3i}{s + i} \\
\Rightarrow y(t) &= 6t + 3ie^{it} - 3ie^{-it} \\
&= 6t - 6 \sin(t)
\end{aligned}$$

(e)

$$\begin{aligned}
s \mathcal{L}[y] - i\omega \mathcal{L}[y] &= (s - i\omega) \mathcal{L}[y] \\
&= 1 \\
\Rightarrow \mathcal{L}[y] &= \frac{1}{s - i\omega} \\
\Rightarrow y(t) &= e^{i\omega t}
\end{aligned}$$

(f)

$$\begin{aligned}
m(s^2 \mathcal{L}[y] - s) + c(s \mathcal{L}[y] - 1) + k \mathcal{L}[y] &= 0 \\
\Rightarrow \mathcal{L}[y] &= \frac{ms + c}{ms^2 + cs + k} \\
&= \frac{s + c/m}{s^2 + (c/m)s + k/m}
\end{aligned}$$

The roots of the denominator are

$$s_{1,2} = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m}$$

The partial fractions decomposition is

$$\mathcal{L}[y] = \frac{s + c/m}{(s - s_1)(s - s_2)} = \left[ \frac{-(c/m + s_1)}{s_2 - s_1} \right] \frac{1}{s - s_1} - \left[ \frac{-(c/m + s_2)}{s_1 - s_2} \right] \frac{1}{s - s_2}$$

By Vieta, we know that  $s_1 + s_2 = -c/m$ . Then we can simplify to

$$\mathcal{L}[y] = -\left( \frac{s_2}{s_1 - s_2} \right) \frac{1}{s - s_1} - \left( \frac{s_1}{s_1 - s_2} \right) \frac{1}{s - s_2}$$

Which corresponds to the  $t$ -domain function

$$y(t) = -\left( \frac{s_2}{s_1 - s_2} \right) e^{s_1 t} - \left( \frac{s_1}{s_1 - s_2} \right) e^{s_2 t}$$

**Question: 8.5.12**

The transform of  $e^{At}$  is  $(sI - A)^{-1}$ . Compute that matrix (the transfer function) when  $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ . Compare the poles of the transform to the eigenvalues of  $A$ .

Compute

$$\begin{aligned} (sI - A)^{-1} &= \begin{bmatrix} s-1 & -1 \\ -1 & s-1 \end{bmatrix}^{-1} \\ &= \frac{1}{(s-1)^2 - 1} \begin{bmatrix} s-1 & 1 \\ 1 & s-1 \end{bmatrix} \end{aligned}$$

Here, we have poles  $s = 0$  and  $s = 2$ . These also turn out to be the eigenvalues.

**Question: 8.5.13**

If  $dy/dt$  decays exponentially, show that  $sY(s) \rightarrow y(0)$  as  $s \rightarrow \infty$ .

We calculate, using integration of parts:

$$\begin{aligned} sY(s) &= \int_0^\infty s e^{-st} y(t) dt \\ &= \int_0^\infty e^{-st} \frac{dy}{dt} dt - e^{-st} y(t) \Big|_0^\infty \\ &= \int_0^\infty e^{-st} \frac{dy}{dt} dt + y(0) \end{aligned}$$

Now, by premise,  $dy/dt$  decays exponentially. Critically, that means it will not “overpower” the  $e^{-st}$  term, which means for large  $s$  the integral converges to 0.

$$\lim_{s \rightarrow \infty} sY(s) = \lim_{s \rightarrow \infty} \int_0^\infty e^{-st} \frac{dy}{dt} dt + y(0) = y(0)$$

The conclusion here is that as  $s$  approaches infinity, the  $s$ -domain transform under these conditions approaches zero.

$$\lim_{s \rightarrow \infty} \mathcal{L}[y'(t)] = \lim_{s \rightarrow \infty} [s\mathcal{L}[y(t)] - y(0)] = 0$$

**Question: 8.5.14**

Transform Bessel’s time-varying equation  $ty'' + y' + ty = 0$  using  $\mathcal{L}[ty] = -dY/ds$  to find a first-order equation for  $Y$ . By separating variables or by substituting  $Y(s) = C/\sqrt{1+s^2}$ , find the Laplace transform of the Bessel function  $y = J_0$ .

Since  $\mathcal{L}[y''(t)] = s^2Y(s) - sy(0) - y'(0)$ , it follows that (using the author’s notation of  $\mathcal{L}[y] = Y(s)$ )

$$\mathcal{L}[ty''(t)] = -2s\mathcal{L}[y] - s^2Y'(s) + y(0)$$

Transforming to the  $s$ -domain (using the author's notation of  $\mathcal{L}[y] = Y(s)$ ), we have

$$(1 + s^2) Y'(s) + sY(s) = 0$$

By separable variables, solve

$$\begin{aligned} \int \frac{dY}{Y} &= - \int \frac{s ds}{1 + s^2} \\ \ln Y(s) &= -\frac{1}{2} \ln(1 + s^2) + C \\ &= \ln \frac{1}{\sqrt{1 + s^2}} + C \\ \Rightarrow Y(s) &= \frac{C}{\sqrt{1 + s^2}} \end{aligned}$$

#### Question: 8.5.15

Find the Laplace transform of a single arch of  $f(t) = \sin(\pi t)$ .

The first arch spans  $0 \leq t \leq 1$ , but let us approach this by considering  $g(t) = e^{i\pi t}$ , and so we derive

$$\begin{aligned} \mathcal{L}[g(t)] &= \int_0^1 e^{-st} e^{i\pi t} dt \\ &= \int_0^1 e^{(-s+i\pi)t} dt \\ &= \left. \frac{e^{(-s+i\pi)t}}{-s+i\pi} \right|_0^1 \\ &= \frac{e^{(-s+i\pi)} - 1}{-s+i\pi} \\ &= \frac{(e^{-s+i\pi} - 1)(-s-i\pi)}{s^2 + \pi^2} \\ &= \frac{-se^{-s+i\pi} + s - i\pi e^{-s+i\pi} + i\pi}{s^2 + \pi^2} \\ &= \frac{se^{-s} + s + i\pi e^{-s} + i\pi}{s^2 + \pi^2} \end{aligned}$$

Taking the imaginary part, we have

$$\mathcal{L}[f(t)] = \frac{\pi(e^{-s} + 1)}{s^2 + \pi^2}$$

**Question: 8.5.16**

Your acceleration  $v' = c(v^* - v)$  depends on the velocity  $v^*$  of the car ahead:

- (a) Find the ratio of Laplace transforms  $V^*(s)/V(s)$ .
- (b) If that car has  $v^* = t$  find your velocity  $v(t)$  starting from  $v(0) = 0$ .

- (a) Rewrite as  $v^* = v'/c + v$  and take the transform to get (assume  $v(0) = 0$ )

$$V^*(s) = \frac{sV(s)}{c} + V(s)$$

Next, divide through by  $V(s)$  to find

$$\frac{V^*(s)}{V(s)} = \frac{s+c}{c}$$

- (b) Rewrite the equation as

$$v' + cv = ct$$

Transforming to the  $s$ -domain, we can derive

$$sV(s) + cV(s) = (s+c)V(s) = \frac{c}{s^2} \implies V(s) = \frac{c}{s^2(s+c)}$$

The partial fraction decomposition gives us

$$V(s) = -\frac{1}{cs} + \frac{1}{s^2} + \frac{1}{c(s+c)}$$

Transforming back to the  $t$ -domain finally offers the solution

$$v(t) = -\frac{1}{c} + t + \frac{1}{c}e^{-ct}$$

**Question: 8.5.17**

A line of cars has  $v'_n = c[v_{n-1}(t-T) - v_n(t-T)]$  with  $v_0(t) = \cos(\omega t)$  in front.

- Find the growth factor  $A = 1/(1 + i\omega e^{i\omega T}/c)$  in oscillation  $v_n = A^n e^{i\omega t}$ .
- Show that  $|A| < 1$  and the amplitudes are safely decreasing if  $cT < \frac{1}{2}$ .
- If  $cT > \frac{1}{2}$  show that  $|A| > 1$  (dangerous) for small  $\omega$ . (Use  $\sin(\theta) < \theta$ .) Human reaction time is  $T \geq 1$  sec and human aggressiveness is  $c = 0.4$  / sec. Danger is pretty close. Probably drivers adjust to be barely safe.

- Using  $v_n = A^n e^{i\omega t}$ , substitute into the equation to find

$$\begin{aligned}
 v'_n &= A^n i\omega e^{i\omega t} \\
 v_{n-1}(t-T) &= A^{n-1} e^{i\omega(t-T)} \\
 v_n(t-T) &= A^n e^{i\omega(t-T)} \\
 A^n i\omega e^{i\omega t} &= c \left[ A^{n-1} e^{i\omega(t-T)} - A^n e^{i\omega(t-T)} \right] \\
 \implies A \left[ i\omega e^{i\omega t} + c e^{i\omega(t-T)} \right] &= c e^{i\omega(t-T)} \\
 \implies A \left[ i\omega + c e^{-i\omega T} \right] &= c e^{-i\omega T} \\
 \implies A &= \frac{c e^{-i\omega T}}{i\omega + c e^{-i\omega T}} \\
 &= \frac{1}{1 + \frac{i\omega e^{i\omega T}}{c}}
 \end{aligned}$$

- There is a great deal of algebra evolved, but at the end of it we find the amplitude

$$|A| = \frac{c^2}{c^2 - 2c\omega \sin(\omega T) + \omega^2}$$

However, we will make the approximation that  $\omega T \ll 1$ , so  $\sin(\omega T) \approx \omega T$ . This allows us the much-needed simplification of

$$|A| \approx \frac{c^2}{c^2 - 2cT\omega^2 + \omega^2}$$

Observe that when  $cT = 1$ , we have  $|A| \approx 1$ . Therefore, we see that  $|A| < 1$  ensues when  $cT < 1/2$ .

- Conversely, when  $cT > 1/2$ ,  $|A| > 1$ .

**Question: 8.5.18**

For  $f(t) = \delta(t)$ , the transform  $F(s) = 1$  is the limit of transforms of tall thin box functions  $b(t)$ . The boxes have width  $\epsilon \rightarrow 0$  and height  $1/\epsilon$  and area 1.

$$\text{Inside integrals, } b(t) = \begin{cases} 1/\epsilon & \text{for } 0 \leq t < \epsilon \\ 0 & \text{otherwise} \end{cases} \text{ approaches } \delta(t)$$

Find the transform  $B(s)$ , depending on  $\epsilon$ . Compute the limit of  $B(s)$  as  $\epsilon \rightarrow 0$ .

Derive

$$\begin{aligned} \mathcal{L}[b(t)] &= \int_0^\epsilon \frac{1}{\epsilon} e^{-st} dt \\ &= -\frac{1}{\epsilon s} e^{-st} \Big|_0^\epsilon \\ &= \frac{1 - e^{-s\epsilon}}{\epsilon s} \end{aligned}$$

Using l'Hôpital's rule, the limit is computed as

$$\lim_{\epsilon \rightarrow 0} \frac{1 - e^{-s\epsilon}}{\epsilon s} = \lim_{\epsilon \rightarrow 0} \frac{\epsilon e^{-s\epsilon}}{\epsilon} = \lim_{\epsilon \rightarrow 0} e^{-s\epsilon} = 1$$

which turns out to be the transform of the delta function, as anticipated.

**Question: 8.5.19**

The transform  $1/s$  of the unit step function  $H(t)$  comes from the limit of the transforms of short steep ramp functions  $r_\epsilon(t)$ . These ramps have slope  $1/\epsilon$ . Compute

$$R_\epsilon(s) = \int_0^\epsilon \frac{t}{\epsilon} e^{-st} dt + \int_\epsilon^\infty e^{-st} dt$$

Let  $\epsilon \rightarrow 0$ .

Compute

$$\begin{aligned} R_\epsilon(s) &= \int_0^\epsilon \frac{t}{\epsilon} e^{-st} dt + \int_\epsilon^\infty e^{-st} dt \\ &= \int_0^\epsilon \frac{e^{-st}}{\epsilon s} dt - \frac{te^{-st}}{\epsilon s} \Big|_0^\epsilon - \frac{e^{-st}}{s} \Big|_\epsilon^\infty \\ &= -\frac{e^{-st}}{\epsilon s^2} \Big|_0^\epsilon - \frac{e^{-s\epsilon}}{s} + \frac{e^{-s\epsilon}}{s} \\ &= \frac{1 - e^{-s\epsilon}}{\epsilon s^2} \end{aligned}$$

By l'Hôpital's rule, we conclude

$$\lim_{\epsilon \rightarrow 0} \frac{1 - e^{-s\epsilon}}{\epsilon s^2} = \lim_{\epsilon \rightarrow 0} \frac{se^{-s\epsilon}}{s^2} = \frac{1}{s}$$

which is the transform of  $H(t)$ .

**Question: 8.5.20**

In Problems 18 and 19, show that the derivative of the ramp function  $r_\epsilon(t)$  is the box function  $b(t)$ . The “generalized derivative” of a step is the \_\_\_\_ function.

The definition of the ramp function is

$$r_\epsilon = \begin{cases} t/\epsilon & \text{for } 0 \leq t < \epsilon \\ 0 & \text{otherwise} \end{cases}$$

Differentiating in each case produces the box function. One issue here is in the “jumps” of the derivative at  $t = 0$  and  $t = \epsilon$ : immediately to the left of  $t = 0$ , the derivative is 0, and to the right,  $1/\epsilon$ . So in the true sense, we do not have differentiability. The notion of a generalized derivative is to extend the idea of differentiability. For more, see distribution theory in real analysis.

**Question: 8.5.21**

What is the Laplace transform of  $y'''(t)$  when you are given  $Y(s)$  and  $y(0)$ ,  $y'(0)$ ,  $y''(0)$ ?

$$\begin{aligned} \mathcal{L}[y'''(t)] &= s\mathcal{L}[y''(t)] - y''(0) \\ &= s^3\mathcal{L}[y] - s^2y(0) - sy'(0) - y''(0) \end{aligned}$$

In general,

$$\mathcal{L}[y^{(n)}(t)] = s^n\mathcal{L}[y] - \sum_{k=0}^{n-1} s^{n-k-1}y^{(k)}(0)$$

**Question: 8.5.22**

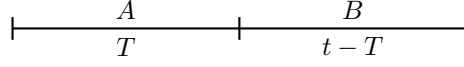
The *Pontryagin maximum principle* says that the optimal control is “bang-bang” – it only takes on the extreme values permitted by the constraints. To go from rest at  $x = 0$  to rest at  $x = 1$  in minimum time, use maximum acceleration  $A$  and deceleration  $-B$ . At what time  $t$  do you change from the accelerator to the brake? (This is the fastest driving between two red lights.)

It is imperative that we are organized with our constraints. They are

$$x_A + x_B = 1, \quad v_{0,A} = v_{f,B} = 0$$

Where  $v_0$  and  $v_f$  correspond to the initial and final velocity, respectively. Now, refer to the below illustration. Let  $T$  be the time we hit the breaks, but it is also the time spent accelerating at  $A$ . And  $t - T$  is the time decelerating at  $-B$ . The total time to get from  $x = 0$  to  $x = 1$  is  $t$ .





There are some immediate physical conclusions we can draw. The first is at the end of section  $A$ , our final velocity is  $v_{f,A} = AT$ , which is also equal to  $v_{0,B}$ . Integrate that to get the distance traversed in that first section, which is  $x_A = \frac{1}{2}AT^2$ .

In the second leg, we can immediately deduce that we travel a distance  $x_B = 1 - \frac{1}{2}AT^2$ . To bring the deceleration into the picture, we can also deduce that  $x_B$  can be written as

$$x_B = AT(t - T) - \frac{1}{2}B(t - T)^2$$

where  $t - T$  corresponds to the time elapsed during the deceleration phase. This equation stems from the one-dimensional kinematics equations, although you can derive it by integrating the acceleration term twice. Combining our two derivations for  $x_B$ , we can write

$$1 = -\frac{1}{2}(A + B)T^2 + T(A + B)t - \frac{1}{2}Bt^2$$

So far, we have one equation and two unknowns. A second equation is needed. We have satisfied the distance constraint, but now we need a *velocity constraint*. Given the initial velocity at the start of section  $B$ , the deceleration constant, and knowledge that the final velocity must be zero, then we can build our second equation as such:

$$AT - B(t - T) = 0$$

Isolating  $t$  gives us

$$t = \frac{1}{B}T(A + B)$$

Finally, we may substitute back into our first equation and solve for the brake time  $T$  to get

$$T = \sqrt{\frac{2B}{A(A + B)}}$$

In short, the kernel of this problem setup is in understanding that we have two unknowns,  $t$  and  $T$ , and thus we will need two constraint equations (distance and velocity) to solve for them.

Note that having the acceleration and deceleration constants given to us as constants simplifies the problem drastically. Otherwise, calculus of variations would have been needed.

## 8.6 - Convolution (Fourier and Laplace)

### Question: 8.6.1

Find the convolution  $\mathbf{v} * \mathbf{w}$  and also the cyclic convolution  $\mathbf{v} \circledast \mathbf{w}$ :

- (a)  $\mathbf{v} = (1, 2)$  and  $\mathbf{w} = (2, 1)$   
 (b)  $\mathbf{v} = (1, 2, 3)$  and  $\mathbf{w} = (4, 5, 6)$

(a)

$$\mathbf{v} * \mathbf{w} = \begin{bmatrix} 1 & & \\ 2 & 1 & \\ & 2 & \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 2 \end{bmatrix}$$

$$\mathbf{v} \circledast \mathbf{w} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

(b)

$$\mathbf{v} * \mathbf{w} = \begin{bmatrix} 1 & & & \\ 2 & 1 & & \\ 3 & 2 & 1 & \\ & 3 & 2 & \\ & & 3 & \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 4 \\ 13 \\ 28 \\ 27 \\ 18 \end{bmatrix}$$

$$\mathbf{v} \circledast \mathbf{w} = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 1 & 3 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 31 \\ 31 \\ 28 \end{bmatrix}$$

As a rule, the length of the convolution of two vectors of length  $n$  is  $2n - 1$ . The cyclic convolution of two vectors of length  $n$  is also length  $n$ .

### Question: 8.6.2

Compute the convolution  $(1, 3, 1) * (2, 2, 3) = (a, b, c, d, e)$ . To check your answer, add  $a + b + c + d + e$ . That total should be 35 since  $1 + 3 + 1 = 5$  and  $2 + 2 + 3 = 7$  and  $5 \times 7 = 35$ .

$$(1, 3, 1) * (2, 2, 3) = \begin{bmatrix} 1 & & & & \\ 3 & 1 & & & \\ 1 & 3 & 1 & & \\ & 1 & 3 & & \\ & & 1 & & \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 8 \\ 11 \\ 11 \\ 3 \end{bmatrix}$$

Confirm that  $2 + 8 + 11 + 11 + 3 = 35$ .

**Question: 8.6.3**

Multiply  $1 + 3x + x^2$  times  $2 + 2x + 3x^2$  to find  $a + bx + cx^2 + dx^3 + ex^4$ . Your multiplication was the same as the convolution  $(1, 3, 1) * (2, 2, 3)$  in Problem 2. When  $x = 1$ , your multiplication shows why  $1 + 3 + 1 = 5$  times  $2 + 2 + 3 = 7$  agrees with  $a + b + c + d + e = 35$ .

$$\begin{aligned}(1 + 3x + x^2) \times (2 + 2x + 3x^2) &= (2 + 2x + 3x^2) + (6x + 6x^2 + 9x^3) + (2x^2 + 2x^3 + 3x^4) \\ &= 2 + 8x + 11x^2 + 11x^3 + 3x^4\end{aligned}$$

**Question: 8.6.4**

(Deconvolution) Which vector  $\mathbf{v}$  would you convolve with  $\mathbf{w} = (1, 2, 3)$  to get  $\mathbf{v} * \mathbf{w} = (0, 1, 2, 3, 0)$ ? Which  $\mathbf{v}$  gives  $\mathbf{v} \circledast \mathbf{w} = (3, 1, 2)$ ?

The identity vector in this instance is  $\mathbf{v} = (0, 1, 0)$ . Confirm by calculating

$$\mathbf{v} * \mathbf{w} = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Observing how the elements of  $\mathbf{w}$  are permuted, we can conclude

$$\mathbf{v} \circledast \mathbf{w} = \begin{bmatrix} & & 1 \\ 1 & & \\ & 1 & \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix} \implies \mathbf{v} = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$$

**Question: 8.6.5**

- For the periodic functions  $f(x) = 4$  and  $g(x) = 2 \cos(x)$ , show that  $f * g$  is **zero** (the zero function)!
- In frequency space ( $k$ -space) you are multiplying the Fourier coefficients of 4 and  $2 \cos(x)$ . Those coefficients are  $c_0 = 4$  and  $d_1 = d_{-1} = 1$ . Therefore every product  $c_k d_k$  is \_\_\_\_.

- (a) Calculate

$$(f * g)(x) = \int_0^{2\pi} g(y) f(x - y) dy = \int_0^{2\pi} 8 \cos(y) dy = 8 \sin(y) \Big|_0^{2\pi} = 0$$

- (b) Every product  $c_k d_k = 0$ .

**Question: 8.6.6**

For periodic functions  $f = \sum c_k e^{ikx}$  and  $g = \sum d_k e^{ikx}$ , the Fourier coefficients of  $f * g$  are  $2\pi c_k d_k$ . Test this factor  $2\pi$  when  $f(x) = 1$  and  $g(x) = 1$  by computing  $f * g$  from its definition (4).

$$(f * g)(x) = \int_0^{2\pi} dy = 2\pi$$

**Question: 8.6.7**

Show by integration that the periodic convolution

$$\int_0^{2\pi} \cos(x) \cos(t - x) dx$$

is  $\pi \cos(t)$ .

In  $k$ -space you are squaring Fourier coefficients  $c_1 = c_{-1} = \frac{1}{2}$  to get  $\frac{1}{4}$  and  $\frac{1}{4}$ ; these are the coefficients of  $\frac{1}{2} \cos(t)$ . The  $2\pi$  in Problem 6 makes  $\pi \cos(t)$  correct.

Calculate

$$\begin{aligned} \int_0^{2\pi} \cos(x) \cos(t - x) dx &= \frac{1}{2} \int_0^{2\pi} [\cos(2x - t) + \cos(t)] dx \\ &= \frac{1}{4} \sin(2x - t) + \frac{1}{2} x \cos(t) \Big|_0^{2\pi} \\ &= \pi \cos(t) \end{aligned}$$

**Question: 8.6.8**

Explain why  $f * g$  is the same as  $g * f$  (periodic or infinite convolution).

One intuitive explanation is to consider two sheets of paper, and overlapping one with the other. It does not matter which sheet is moved to overlap with the other; the overlapped area will be the same irrespective of that choice.

Mathematically, a simple variable substitution  $\tau = x - y$  allows one to switch between  $f * g$  or  $g * f$ . Additionally, convolution in function space corresponds to multiplication in frequency space, and since the product of coefficients commutes, it follows that convolution also commutes.

**Question: 8.6.9**

What 3 by 3 circulant matrix  $C$  produces cyclic convolution with the vector  $\mathbf{c} = (1, 2, 3)$ ? Then  $C\mathbf{d}$  equals  $\mathbf{c} \circledast \mathbf{d}$  for every vector  $\mathbf{d}$ . Compute  $\mathbf{c} \circledast \mathbf{d}$  for  $\mathbf{d} = (0, 1, 0)$ .

$$C = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 1 & 3 \\ 3 & 2 & 1 \end{bmatrix} \implies C\mathbf{d} = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 1 & 3 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix} = \mathbf{c} \circledast \mathbf{d}$$

**Question: 8.6.10**

What 2 by 2 circulant matrix  $C$  produces cyclic convolution with  $\mathbf{c} = (1, 1)$ ? Show in four ways that this  $C$  is not invertible. Deconvolution is impossible.

- (1) Find the determinant of  $C$ .
- (2) Find the eigenvalues of  $C$ .
- (3) Find  $\mathbf{d}$  so that  $C\mathbf{d} = \mathbf{c} \circledast \mathbf{d}$  is zero.
- (4)  $F\mathbf{c}$  has a zero component.

The circulant matrix is

$$C = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

- (1)  $\det C = 1 - 1 = 0$ , zero determinant implies non-invertibility.

(2)

$$\det \begin{bmatrix} 1 - \lambda & 1 \\ 1 & 1 - \lambda \end{bmatrix} = (1 - \lambda)^2 - 1 = 0 \implies \lambda_{1,2} = 2, 0$$

A zero eigenvalue implies non-invertibility.

- (3) Observe that

$$\mathbf{d} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \implies C\mathbf{d} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \mathbf{c} \circledast \mathbf{d}$$

which implies linear dependence in the rows and thus non-invertibility.

(4)

$$F\mathbf{c} = \begin{bmatrix} 1 & 1 \\ 1 & w \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 + w \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

where  $w = e^{-\pi i}$ . Observe that  $\mathbf{c}$  is one of the eigenvectors (corresponding to eigenvalue  $\lambda = 2$ ), and the Fourier matrix times that eigenvector restores the eigenvalues.

**Question: 8.6.11**

- (a) Change  $b(x) * \delta(x - 1)$  to a multiplication  $\hat{b}\hat{d}$ . Transform the box function

$$b(x) = \{1 \text{ for } 0 \leq x \leq 1\} \text{ to } \hat{b}(k) = \int_0^1 e^{-ikx} dx$$

The shifted delta transforms to

$$\hat{d}(k) = \int \delta(x - 1) e^{-ikx} dx$$

- (b) Show that your result  $\hat{b}\hat{d}$  is the transform of a shifted box function. Then convolution with  $\delta(x - 1)$  shifts the box.

- (a) Convolution without working in frequency space gives us

$$b(x) * \delta(x - 1) = \int_{-\infty}^{\infty} b(y) \delta(x - 1 - y) dy = b(x - 1) = \begin{cases} 1, & 1 \leq x \leq 2 \\ 0, & \text{else} \end{cases}$$

which corresponds to a one-step shift in  $b(x)$ . Meanwhile, multiplication in frequency space gives

$$\hat{b}\hat{d} = \int_0^1 e^{-ikx} dx \int_{-\infty}^{\infty} \delta(x - 1) e^{-ikx} dx = \left[ -\frac{e^{-ikx}}{ik} \Big|_0^1 \right] e^{-ik} = \left[ \frac{1 - e^{-ik}}{ik} \right] e^{-ik}$$

- (b) The transform of the one-step shifted  $b(x)$  is

$$\hat{b}(k) = \int_1^2 e^{-ikx} dx = \left[ \frac{1 - e^{-ik}}{ik} \right] e^{-ik}$$

**Question: 8.6.12**

Take the Laplace transform of these equations to find the transfer function  $G(s)$ :

- (a)  $Ay'' + By' + Cy = \delta(t)$   
 (b)  $y' - 5y = \delta(t)$   
 (c)  $2y(t) - y(t - 1) = \delta(t)$

Assume zero for all initial conditions.

- (a)  $As^2Y + BsY + CY = 1 \implies Y(s) = \frac{1}{As^2 + Bs + C}$   
 (b)  $sY - 5Y = 1 \implies Y(s) = \frac{1}{s - 5}$   
 (c)  $2Y - Ye^{-s} = 1 \implies Y(s) = \frac{1}{2 - e^{-s}}$

**Question: 8.6.13**

Take the Laplace transform of  $y'''' = \delta(t)$  to find  $Y(s)$ . From the Transform Table in Section 8.5 find  $y(t)$ . You will see  $y''' = 1$  and  $y'''' = 0$ . But  $y(t) = 0$  for negative  $t$ , so your  $y'''$  is actually a unit step function and your  $y''''$  is actually  $\delta(t)$ .

We have  $s^4 Y = 1$  which gives us

$$Y(s) = \frac{1}{s^4}$$

This corresponds to  $y(t) = t^3/6$ . By the discontinuity at  $t = 0$  since  $y(t) = 0$  for negative  $t$ , we must have  $y'''' = \delta(t)$  and not  $y'''' = 0$ .

**Question: 8.6.14**

Solve these equations by Laplace transform to find  $Y(s)$ . Invert that transform with the Table in Section 8.5 to recognize  $y(t)$ .

(a)  $y' - 6y = e^{-t}, y(0) = 2$

(b)  $y'' + 9y = 1, y(0) = y'(0) = 0$

$$(a) \quad sY - 2 - 6Y = \frac{1}{s+1} \quad \Rightarrow \quad Y(s) = \frac{1}{(s+1)(s-6)} + \frac{2}{s-6}$$

By partial fractions,

$$Y(s) = -\frac{1}{7(s+1)} + \frac{15}{7(s-6)}$$

which corresponds to time domain solution

$$y(t) = \frac{15}{7}e^{6t} - \frac{1}{5}e^{-t}$$

$$(b) \quad s^2 Y + 9Y = \frac{1}{s} \quad \Rightarrow \quad Y(s) = \frac{1}{s(s^2 + 9)}$$

By partial fractions,

$$Y(s) = \frac{1}{3s} - \frac{s}{3(s^2 + 3)}$$

which corresponds to time domain solution

$$y(t) = \frac{1}{9} - \frac{1}{9} \cos(3t)$$

**Question: 8.6.15**

Find the Laplace transform of the shifted step  $H(t - 3)$  that jumps from 0 to 1 at  $t = 3$ . Solve  $y' - ay = H(t - 3)$  with  $y(0) = 0$  by finding the Laplace transform  $Y(s)$  and then its inverse transform  $y(t)$ : one part for  $t < 3$ , second part for  $t \geq 3$ .

$$sY - aY = \frac{e^{-3s}}{s} \implies Y(s) = \frac{e^{-3s}}{s(s-a)} = e^{-3s} \left[ -\frac{1}{as} + \frac{1}{a(s-a)} \right]$$

which corresponds to time domain solution

$$y(t) = \frac{1}{a} \left[ e^{a(t-3)} - 1 \right] H(t-3)$$

**Question: 8.6.16**

Solve  $y' = 1$  with  $y(0) = 4$ —a trivial question. Then solve this problem the slow way by finding  $Y(s)$  and inverting that transform.

The solution can be derived on inspection:  $y(t) = t + 4$ . By transforms, we have

$$sY - 4 = \frac{1}{s} \implies Y(s) = \frac{1}{s^2} + \frac{4}{s}$$

which has time domain solution (as expected)

$$y(t) = t + 4$$

**Question: 8.6.17**

The solution  $y(t)$  is the convolution of the input  $f(t)$  with what function  $g(t)$ ?

(a)  $y' - ay = f(t)$  with  $y(0) = 3$

(b)  $y' - (\text{integral of } y) = f(t)$

The task is to identify the transfer function  $G(s)$  and its corresponding Green's function  $g(t)$ .

(a)  $sY - 3 - aY = F(s) \implies Y(s) = \frac{F(s) + 3}{s - a}$  which has time domain solution

$$y(t) = f(t) * e^{at} + 3e^{at}$$

Appreciate the elegance of this operation: even in ignorance of  $f(t)$ , we may still write a solution  $y(t)$  thanks to the switch between the  $s$ - and  $t$ -domains corresponding to a switch between multiplication and convolution.



(b) Two ways to solve: first, attempt to derive the Laplace transform of  $\int y \, dt$ :

$$\mathcal{L}\left[\int y(T) \, dT\right] = \int_0^\infty \left[\int_0^t y(T) \, dT\right] e^{-st} \, dt$$

Solve by integration by parts. Let  $u = \int_0^t y(T) \, dT$  and  $dv = e^{-st} \, dt$ . Then  $du = y(t)$  and  $v = -e^{-st}/s$ . We have

$$\begin{aligned} \mathcal{L}\left[\int y(T) \, dT\right] &= -\frac{e^{-st}}{s} \int_0^t y(T) \, dT \Big|_0^\infty + \int_0^\infty y(t) \frac{e^{-st}}{s} \, dt \\ &= \frac{Y(s)}{s} \end{aligned}$$

A convenient result: differentiation produces  $s$ , and integration gives us  $1/s$ . The rest is straightforward.

$$sY - \frac{Y}{s} = F(s) \quad \implies \quad Y(s) = \frac{F(s)}{s - \frac{1}{s}}, \quad G(s) = \frac{1}{s - \frac{1}{s}}$$

It is clearer if we rewrite  $G(s)$  as

$$G(s) = \frac{s}{s^2 - 1}$$

which is the transform of  $g(t) = \cos(it)$ . Alternatively, we can differentiate through to annihilate the integral. Equivalently, we solve

$$y'' - y = f'(t)$$

Applying the transform gets us

$$s^2Y - Y = sF(s) \quad \implies \quad Y(s) = \frac{sF(s)}{s^2 - 1}$$

Either way, we derive  $y(t) = f(t) * \cos(it)$ .

#### Question: 8.6.18

For  $y' - ay = f(t)$  with  $y(0) = 3$ , we could replace that initial value by adding  $3\delta(t)$  to the forcing function  $f(t)$ . Explain that sentence.

In the  $s$ -domain, the transform of  $y'$  gets us  $sY - y(0)$ . This produces 3 on the right-hand side. Equivalently, the transform of  $3\delta(t)$  is 3, landing us in the same place.

**Question: 8.6.19**

What is  $\delta(t) * \delta(t)$ ? What is  $\delta(t-1) * \delta(t-2)$ ? What is  $\delta(t-1)$  times  $\delta(t-2)$ ?

Equivalently, in  $s$ -space we have  $1 \cdot 1 = 1$ , which maps back to  $\delta(t)$  in  $t$ -space. In the second equation, we have  $e^{-s} \cdot e^{-2s} = e^{-3s}$  in  $s$ -space, or  $\delta(t-3)$  in  $t$ -space. Lastly,  $\delta(t-1)\delta(t-2) = 0$ , as there is no  $t$  that satisfies a non-zero product of these delta functions.

**Question: 8.6.20**

By Laplace transform, solve  $y' = y$  with  $y(0) = 1$  to find a very familiar  $y(t)$ .

$$sY - 1 = Y \quad \implies \quad Y(s) = \frac{1}{s-1} \quad \implies \quad y(t) = e^t$$

**Question: 8.6.21**

By Fourier transform as in (9), solve  $-y'' + y = \text{box function } b(x)$  on  $0 \leq x \leq 1$ .

Derive

$$-(ik)^2 \hat{y} + \hat{y} = \int_0^1 e^{-ikx} dx = \frac{1 - e^{-ik}}{ik} \quad \implies \quad \hat{y} = \frac{1 - e^{-ik}}{ik(k^2 + 1)}$$

Decomposing  $\hat{y}$  into  $F(s)G(s)$ , we have

$$\begin{aligned} \hat{y} &= \left( \frac{1 - e^{-ik}}{ik} \right) \left( \frac{1}{k^2 + 1} \right) \quad \implies \quad y(x) = [H(x) - H(x-1)] * \frac{1}{2} e^{-|x|} \\ &= \frac{1}{2} \int_0^1 e^{-|x-y|} dy \end{aligned}$$

This is the transform method for solving this equation. The alternative is to solve the equation piecewise, and deriving the coefficients from the junctions at  $x = 0$  and  $x = 1$ . For students of quantum mechanics, you may find this reminiscent of solving the wave equation in “particle-in-a-box” scenarios.

**Question: 8.6.22**

There is a big difference in the solutions to  $y'' + By' + Cy = f(x)$ , between the cases  $B^2 < 4C$  and  $B^2 > 4C$ . Solve  $y'' + y = \delta$  and  $y'' - y = \delta$  with  $y(\pm\infty) = 0$ .

Since we look at the full real line, we apply the Fourier transform to derive

$$(ik)^2 \hat{y} + \hat{y} = 1 \quad \implies \quad \hat{y} = \frac{1}{1 - k^2}$$

Taking the inverse Fourier transform gives

$$y(x) = \frac{1}{2} \sin(|x|)$$

For this periodic function, we do not impose  $y(\pm\infty) = 0$  at the extremes.

In the second question, derive

$$(ik)^2 \hat{y} - \hat{y} = 1 \quad \implies \quad \hat{y} = -\frac{1}{k^2 + 1}$$

which corresponds to  $x$ -domain function

$$y(x) = -\frac{1}{2} e^{-|x|}$$

and in this case, we see that the boundary conditions  $y(\pm\infty) = 0$  hold.

**Question: 8.6.23**

(Review) Why do the constant  $f(t) = 1$  and the unit step  $H(t)$  have the same Laplace transform  $1/s$ ? Answer: Because the transform does not notice \_\_\_\_.

The Laplace transform does not notice  $f(t)$  for  $t < 0$ , i.e.  $f(t) = 0$  on that domain.